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Review

Human-robot collaborative disassembly in Industry 5.0: A systematic literature review and future research agenda

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ABSTRACT

Collaborative human-robot units have attracted recognition for their ability to be used for flexible product disassembly to help achieve intelligent, sustainable, and service-oriented remanufacturing. The adoption of human-robot collaborative disassembly (HRCD) in Industry 5.0 contributes to enhancing the flexibility of the eco-friendly sustainable manufacturing supply chain, realising a circular product life cycle, and facilitating the transition to carbon neutrality. To conduct a systematic examination of the development and research trends in HRCD, a quantitative analysis was carried out on 99 studies retrieved from databases. The research topic structure was examined from an array of perspectives, shedding light on the current state, future pathways, and focal areas in conjunction with a visually depicted knowledge graph. This paper enables scholars to comprehend the trajectory and pivotal aspects of HRCD through investigations of intelligent remanufacturing, thereby clarifying the path for further advancements in sustainable manufacturing practices.

1. Introduction

To achieve artificial intelligence in manufacturing systems while maintaining trust and safety, the European Union has proposed the Industry 5.0 strategy. This pioneering strategy is designed to establish a sustainable, human-centred, and resilient vision for industrial advancement [1–3]. Industry 5.0 (I5.0) signifies a paradigm shift driven by technology, placing humans at the forefront of integration within cyber-physical systems [4–6]. With a deliberate emphasis on empowering the human element within the system, the integration of human cognitive capacities into subsystems is activated as necessary, thereby establishing an intricate human-cyber-physical system (HCPS) [7]. This not only elevates human well-being but also enhances their capability for independent decision-making, cultivating systemic resilience in the manufacturing process and promoting sustained involvement in the production cycle [8,9]. The amalgamation of remanufacturing technology with I5.0 increases ecological sustainability and amplifies the adaptability of manufacturing supply chains [10,11]. Considering the global limitations on resource availability, influenced by both environmental conservation and economic motivations, remanufacturing has garnered significant attention within the framework of I5.0's sustainability initiatives.

The increasing demand for customised products, the expansion of amenities, and the accelerated process of product replacements have resulted in a proliferation of end-of-life (EOL) products. These EOL products retain significant added value [12], offer substantial recovery potential [13], span diverse fields and cover a wide range of categories [14]. The direct waste of resources and environmental degradation stemming from discarding these products are concerns within the realm of sustainable development. Remanufacturing is a systematic process that allows EOL products to regain their vitality, including disassembly, reprocessing and reassembly. Disassembly, as the initial phase of remanufacturing process, is a prerequisite for efficient product recycling and a key step in realizing resource reuse [15,16]. Manual disassembly is a traditional method for disposing of EOL products ecologically while simultaneously recovering or repurposing their components [17]. However, manual disassembly has proven to be economically burdensome, inefficient, and laborious when faced with high-load and high-risk tasks (e.g., oversized parts, noxious elements, hazardous components) [18]. On the other hand, robot-assisted disassembly excels in terms of efficiency but is inflexible due to its inability to contend with unpredictable variables (e.g., product condition [19], material properties [20], component rigidity [21], external disturbances [22]). As such, human–robot collaborative disassembly (HRCD) seamlessly integrates robots with robust load-bearing capacity, high accuracy, stable

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Nomenclature

AI	artificial intelligence
AR	augmented reality
CNN	convolutional neural networks
DC	disassembly workcell
DE	disassembly evaluation
DLB	disassembly line balancing
DT	digital twins
DSP	disassembly sequence planning
EOL	end of life
FVP	fuel vehicle parts
HRC	human-robot collaboration
HRCD	human-robot collaborative disassembly
HCPS	human-cyber-physical system
I5.0	Industry 5.0
LIB	lithium-ion batteries
MR	mixed reality
NSF	national science foundation
RL	reinforcement learning
RNN	recurrent neural networks
WEEE	waste electrical and electronic equipment

repetitive execution, and the agility and adaptability that characterise human subjective experience and tacit knowledge when faced with uncertain tasks [23,24]. This synergistic combination leads to disassembly systems that are not only adaptable, flexible, and resilient but also possess significantly enhanced ergonomic efficiency. To achieve friendly and efficiently, it is necessary to analyze the interaction mode between human and robot, as well as understand the specific interaction requirements at different links. Therefore, HRCD plays a crucial role in embodying the core values of Industry 5.0.

Owing to the diverse array of disassembly requirements for a multitude of products, stringent safety regulations, and intricate working environment, the process of HRCD is plagued with numerous uncertainties. These uncertainties encompass fluctuations in disassembly speed, risks linked to disassembly failures, the quality of EOL products, and factors pertaining to human beings and robot status [25]. Although human beings undoubtedly serve as the most active and dynamic driving force behind HRCD activities, achieving optimal alignment between humans, information systems, and physical infrastructure has proved challenging without acknowledging the pivotal impact of human factors on dismantling operations [26]. The existing research on HRCD primarily centres on the qualitative and quantitative analysis of static product models and data through preprogramming methodologies. The objective is to align and reconcile the economic, social, environmental, and safety considerations within the HRCD process. However, the introduction of variability and uncertainty presents a new set of challenges to collaborative disassembly between humans and robots. Consequently, the focal point of technological advancement lies in establishing security and trust in task assignment as well as human-robot interaction.

Hjorth et al., 2022 [27] conducted a review of HRCD, but their analysis was confined to an in-depth examination of the contents of only 9 articles. In light of the burgeoning interest in HRCD within the academic literature, there is an urgent need for a thorough examination of existing research to discern future research trends and hotspots. This work endeavours to bridge the research gap by combining a review with a forward-looking perspective. In this paper, we present a comprehensive review of the existing publications applying HRCD technologies to EOL recycling. The objective of this work is to provide responses to the following research questions.

- What are the primary themes and presentations in HRCD?
- What are the challenges and barriers of HRCD problem?
- What are the potential HRCD directions and trends to be anticipated toward I5.0?

This study will help foster a detailed understanding of the scope, methodology, and focal points of collaborative disassembly research thus facilitating future enquiries into this topic. This investigation entails exhaustive scrutiny of the literature on HRCD. Our analysis encompasses a thorough examination of 99 studies on HRCD published between 2010 and 2024. This paper is organised as follows (in Fig. 1). Section 2 describes the methodology employed in this review. Section 3 discusses the results of the bibliometric analysis. Section 4 offers an in-depth analysis of the identified studies. Section 5 discusses areas necessitating further research and provides recommendations grounded in the findings. Section 6 concludes the paper.

2. Research methodology

The methodology of bibliometrics possesses the virtues of quantitative assessment, transparent methodology, and robust scalability. This capability enables in-depth analysis of the evolutionary trajectory and developmental trends within a specific field or industry [28,29]. Furthermore, it offers innovative perspectives for information management, the governance of science and technology, and engineering research and related domains [30,31]. CiteSpace is a sophisticated, multi-dimensional dynamic visual tool for conducting knowledge graph analysis. It is utilised to scrutinise the structure, patterns, and distribution characteristics of scientific knowledge [32–34]. Leveraging the findings from keyword time series clustering, co-occurrence, and frequency analyses provides a visually stunning portrayal of network coverage, density structure, evolution, and collaboration within the domain of knowledge. This study employs a systematic analytical approach that integrates bibliometric methodologies and knowledge graphs to perform time-sequential and multi-dimensional text mining, as well as citation analysis, on literature within the realm of HRCD. Furthermore, network co-occurrence and cluster analyses, in conjunction with visual knowledge graphs, are executed to unveil research hotspots and cutting-edge trends in this field.

The dependability and caliber of bibliometric analysis are directly impacted by the data sources employed. The Web of Science, Scopus, and Google Scholar platforms are widely utilised databases that host a plethora of scholarly resources. These three core collection platforms were chosen for this study to conduct advanced data retrieval focused on the theme of “human-robot collaborative disassembly”. The designated time frame for retrieval extends from January 2010 to August 2024. After numerous attempts using a variety of keyword combinations on the databases, the compilation of words presented in Table 1 proves to be apt for carrying out a wide-ranging and inclusive. These identified keywords were systematically categorised into five distinct groups, aligning perfectly with the specific objectives delineated for the literature review. The selection of these terms is predicated on the notion that the documents must bear significance to the collaborative disassembly of robots, recognising that this process relies on the synergy between humans and their robotic counterparts. This meticulous search method, encompassing authorship, title, keywords, and abstracts, resulted in a total of 177 pertinent documents. We exported the results to Rayyan [35], an online tool to aid systematic reviews, by abstract review, excluding 57 papers not related to disassembly problem. This operation allowed excluding the majority of false positives that had arisen from keywords search results. In addition, 21 further papers were excluded, by in depth read, for the following reasons: 18 papers were not in the human-robot or robotic disassembly domain, 3 papers were not accessible. Thus, we successfully reviewed and included into the results of this systematic review a total of 99 papers.

Fig. 2 presents a visual representation of the process of refining the

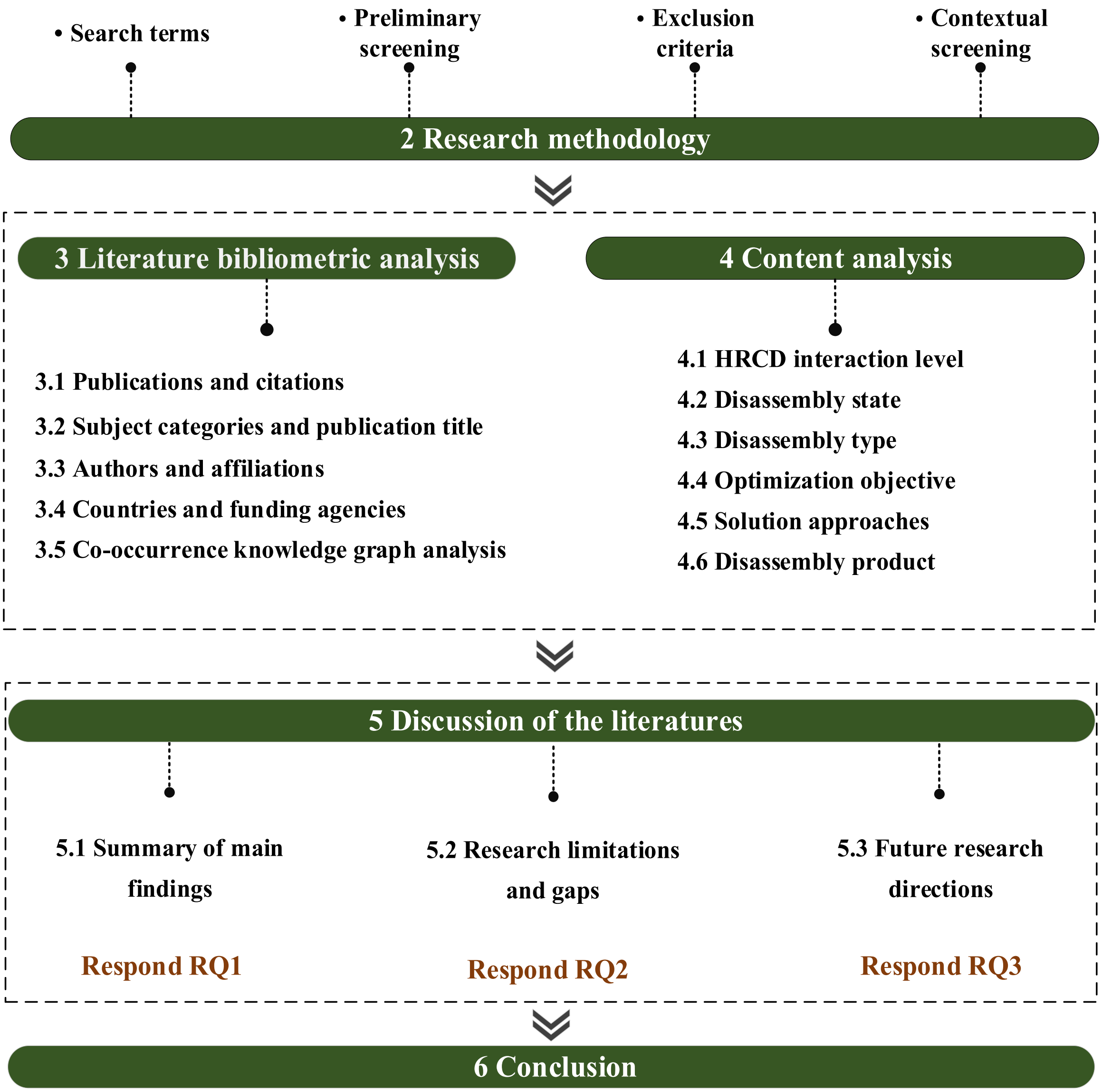

Fig. 1. Structure of this paper.

Table 1
Overview of the various review criteria applied during the process for relevant literature.

Category	Search terms	Preliminary screening	Exclusion criteria	Contextual screening
Description	Human-robot collaborat* disassemb*Robot assisted disassemb*Human machine disassemb*Human-cobot collaborat* disassemb*Cobot assisted disassemb*Robot* disassembly	Time period: January 2010 to August 2024Language: EnglishDocument type: article, review, online publication, conference review paper	Medical scienceBiologyChemicalThermodynamicsArchitectonicPhysicsInstruments instrumentation	Robot programmingStructure and tool designPolicy discourseCommunication and signal transmission Installation operation

pertinent literature. Initially, a total of 177 documents underwent preliminary screening based on criteria such as time interval, language and file type. The search time is initiated in 2010 due to the absence of any pertinent published articles up to that point. The publication language of the article was exclusively English, with the deliberate exclusion of Chinese, French, Japanese and other linguistic variations. The chosen document types encompass papers, reviews, conference papers, and online publications while deliberately excluding patents, dissertations, early access materials and awarded grants. Second, the current study delves into the expansion of HRCD within the field of engineering and its various applications. Areas previously considered beyond the scope of this research encompass disciplines such as medical science, biology, chemical processes, thermodynamics, architecture, physics, instruments and instrumentation. The selection of papers is executed based on a

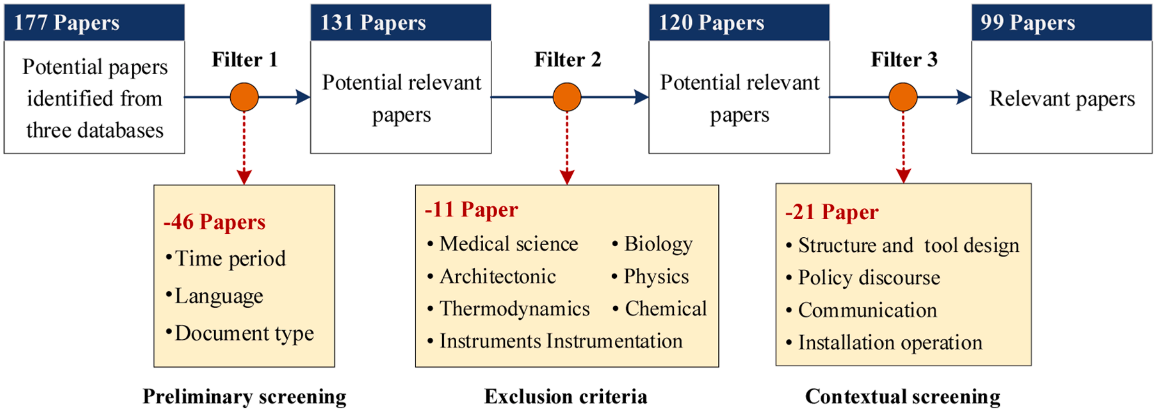

Fig. 2. Procedure for the selection of review articles.

rigorous examination of their titles and abstracts in accordance with the criteria outlined in Table 1, resulting in a total of 177 documents being included. Ultimately, all documents are contextualised through the process of reading and analysing both titles and abstracts. Additionally, any references to robotic teams or fully automated systems without human involvement were omitted from this study. The primary focus of most enterprises lies in the economic feasibility and program feasibility of HRCD itself; thus, papers addressing HRCD policy discourse have been disregarded. Furthermore, the literature pertaining to the design and structure of tools, as well as the installation operation, has been omitted from the selection of papers. Similarly, literature on communication and signal transmission has been excluded due to their general applicability across HRCD applications. Consequently, a total of 99 files were ultimately selected from consideration.

3. Literature bibliometric analysis

3.1. Publications and citations

The temporal distribution of the quantity of published papers functions as a pivotal gauge for monitoring the progress and trajectory of research within a specific field. As illustrated in Fig. 3, the overall path of HRCD research from 2010 to 2024 displays an upwards momentum. While the annual volume of published papers is undergoing rapid expansion, the collective number remains relatively restrained, indicating that HRCD research is still in its nascent stage. In general, the number of papers published in HRCD can be categorised into three distinct stages. During the period from 2010 to 2018, the publication count remained within a modest single-digit range. The subsequent stage, spanning from 2019 to 2022, witnessed a surge in scholarly interest in HRCD, as indicated by a significant increase in annual publication numbers, which soared above single digits and culminated in rapid growth. It is worth noting that there is a small decline in the number of publications in 2020 and 2021, which may be due to the experimental discontinuity caused by COVID-19. After 2023, there was a rapid increase in the number of published papers, with a peak of 67 papers accounting for 39.4 % of the total. Overall, over the past three years, the number of articles accounts for more than half of the total, indicating that HRCD is a prominent research focus. Based on the current trend, it is projected that the number of articles published in 2024 will reach a record high.

In a similar vein, the trajectory of citations can be broadly categorised into three distinct phases: the period of stagnation, the phase of gradual expansion, and the stage of exponential growth. As depicted in Fig. 3, the annual average number of citations remained steady from 2010 to 2018. However, from 2019 to 2023, the annual average number of citations notably increased to 62.5, signifying a period of gradual growth. Over the past three years, there has been a rapid surge in the number of citations, leading to an impressive annual average of 326. In addition, there has been a growing interest in the exploration and dissemination of HRCD over the past 3 years. This trend also underscores the potential for further exploration and study in the future.

3.2. Subject categories and countries

Fig. 4 illustrates the top ten research fields and their respective distributions. Based on the statistical data depicted in Fig. 4, it is evident that engineering has garnered the highest number of academic

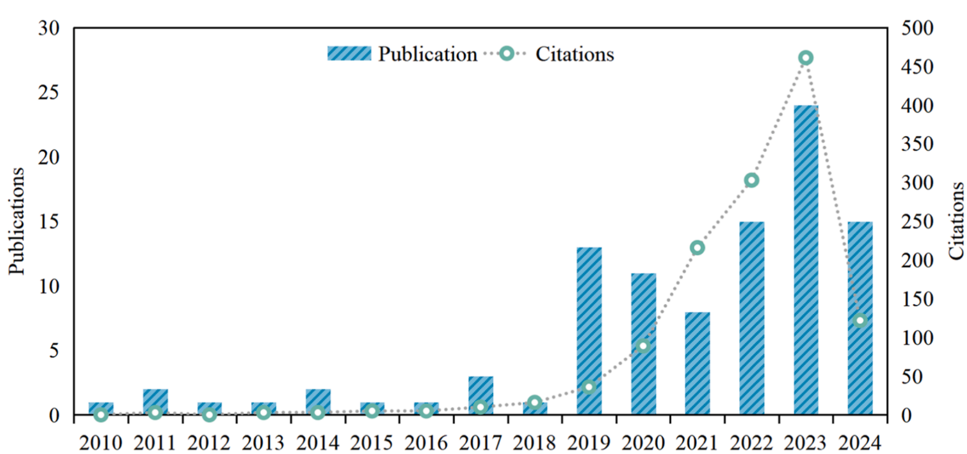

Fig. 3. Annual number of publications and citations.

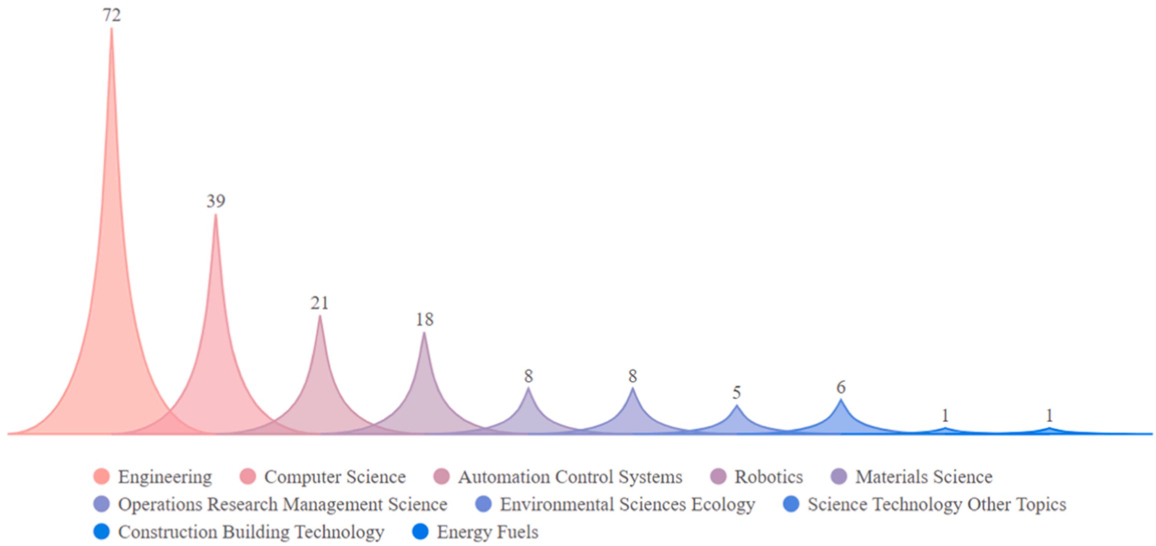

Fig. 4. Distribution of subject categories.

publications, with a total of 72 papers. The next most common are computer science with 39, automation control systems and robotics, with 21 and 18 articles respectively. Furthermore, research on HRCD has extended to the fields of operations research management science, environment sciences ecology, and various other disciplines. Note that there is crossover between disciplines, that is, an article may count more than one discipline. The aforementioned statistical findings support the maturation and widespread application of intelligent manufacturing technology. It is evident that the study of HRCD displays notable interdisciplinarity, encompassing a diverse array of fields as this innovative technology continues to evolve.

An analysis of the countries with published papers, as illustrated in Fig. 5, revealed that China has solidified its position at the pinnacle with 38 impressive published papers. The United Kingdom (UK) and the United States (USA) have both garnered a commendable second place, each boasting 18 published papers. Germany, Spain, Denmark, and various other countries closely follows. On the whole, primary HRCD research is predominantly focused in China, with sizable lead over the combined efforts of the UK and the USA, thus holding an unparalleled advantage. The UK and the USA are the secondary contributors to HRCD research, surpassing their counterparts in Western European nations. Other countries are primarily situated across Western Europe, South Asia and Southeast Asia. It is worth noting that while the USA and Romania have published a greater number of papers, there remains an absence of international collaboration, as they are more inclined towards partnering with their respective national institutions.

3.3. Co-occurrence knowledge graph analysis

By analysing the co-occurrence knowledge graph of subject words, the intricate network of cooperation and relationships between subjects within the realm of research is vividly elucidated. Parameter setting for CiteSpace: establish the temporal scope as 2010–2024, designate the time slice as one year, and impose a threshold of 25. The designated analysis components encompassed the author, keyword, institution, and country categories. The node threshold is determined by selecting the most frequently occurring value within each temporal segment. The graph visualises the relative size of nodes as an indication of the volume

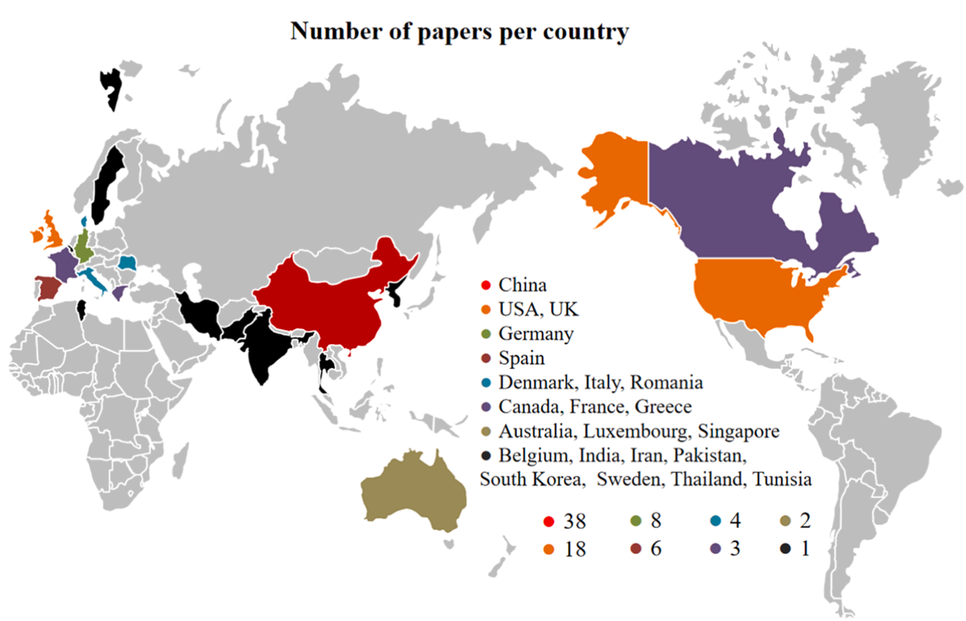

Fig. 5. The map of research distribution all over the world.

of published documents, while the thickness of node connections reflects the intensity of collaboration. Additionally, the annual rings surrounding each node serve to denote the chronological order in which these publications are released.

3.3.1. Author co-occurrence knowledge graph

The construction of author co-occurrence knowledge maps unveils the intricate interconnections among authors and coauthors within the realm of literature. By delineating the author co-occurrence map, we can dissect and analyse the pivotal teams in the domain of HRCD research while shedding light on the epistemological trajectory of research evolution. This study produces a knowledge graph of author co-occurrence, extracted from the analysis of author data, as depicted in Fig. 6. The graph comprises a total of 211 nodes and 536 edges, indicating the interconnectedness and relationships among authors. Note that authors with fewer than 3 publications are not shown in Fig. 6. It shows that there are numerous co-occurrence relationships among the nodes, indicating the initial formation of multiple HRCD research teams and the establishment of certain cooperative networks among some of these teams.

In Fig. 6, the predominant figures in terms of node size are Pham and Zhou, both active in research into HRCD disassembly sequence planning, with a collective publication record of 17 papers. This duo is further supported by a dedicated team including Q Liu, Xu, Fang, Ji, Su, Huang, Wang, J Liu and other esteemed members. The focal point of another team revolves around Behdad, with coauthors including Liang, M Zheng, Lee and others. A third hot spot includes Filipescu, Guo, P Zheng, Xiao, Zhang, and others. Overall, the principal researchers have engaged in extensive studies of HRCD, and their scholarly efforts intricately delineate the research emphases and frontiers within the realm of HRCD.

3.3.2. Affiliations co-occurrence knowledge graph

To demonstrate the interconnectedness of collaborative research across diverse institutions, a co-occurrence knowledge graph depicting HRCD research institutions is depicted in Fig. 7. The graph consists of a total of 85 nodes and 103 edges, illustrating the intricate network of connections among these entities. Overall, the dispersion of nodes exhibits a relatively scattered pattern, with only a handful of nodes being tightly clustered. Collaboration among academic institutions fosters the dissemination of knowledge and the advancement of research in the field of HRCD, both in terms of scope and depth. Notably, multiple individual nodes are dispersed along the periphery of Fig. 7, indicating a palpable fervor in current HRCD research characterised by a vast network of research institutions, numerous high-impact teams, and intimate collaboration. Hence, it is anticipated that scholars will continue further research and exploration into the realm of HRCD. The most prestigious academic establishments include Wuhan University of Technology and the University of Birmingham, with close affiliations to seven institutions, such as the University of Shanghai for Science & Technology, Southeast University, and Hong Kong Polytechnic University. The State University of New York System ranks second in the frequency of hot spots, with close collaboration between the University of Florida. Their cooperative efforts have elevated their position as one of the leading institutions in this regard. In addition, institutions with a high frequency of occurrence include Liaoning Petrochemical University, Southwest Jiaotong University, Wayne State University, and the Centre for Research & Technology Hellas. These partnerships facilitate a robust network of collaboration and innovation at an international level, enriching the academic experience for students and faculty alike.

3.3.3. Keywords co-occurrence knowledge graph

Keywords serve as the dense and concentrated essence of literature research, encapsulating the forefront of cutting-edge research topics. Thus, by conducting statistical analysis on the frequency of these keywords, we are able to discern the pivotal themes within this realm of research. Keyword co-occurrence analysis was conducted using CiteSpace, and the findings are illustrated in Fig. 8. Notably, the primary

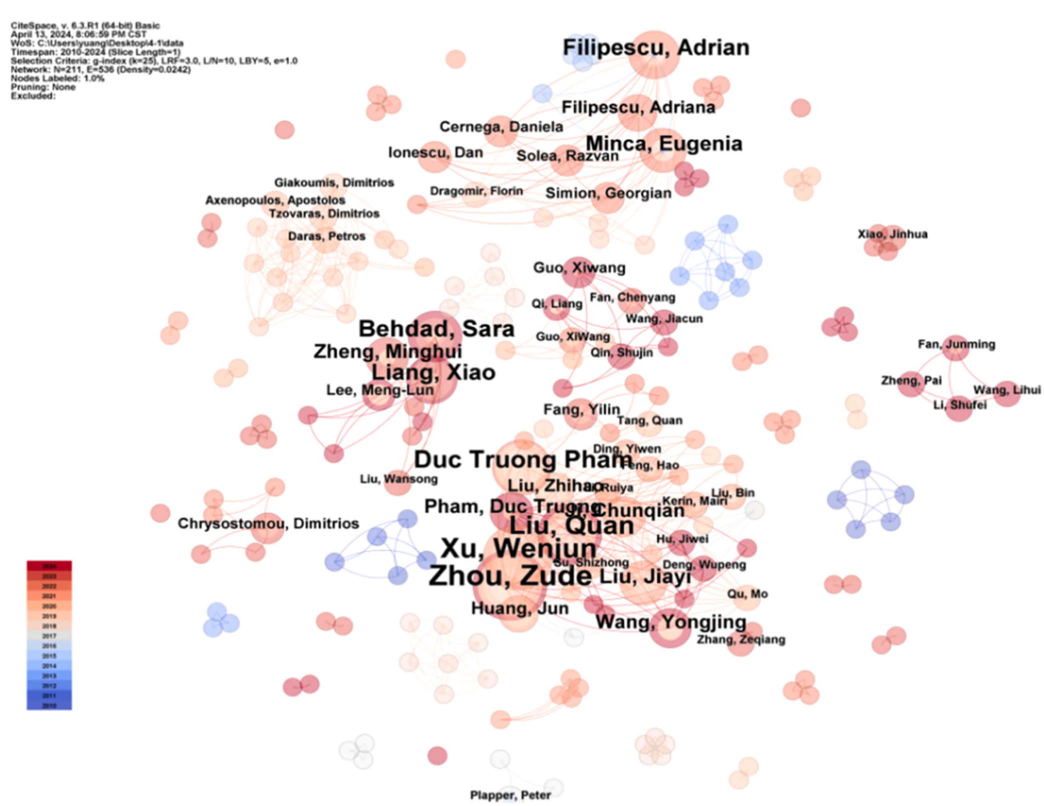

Fig. 6. Author co-occurrence knowledge graph.

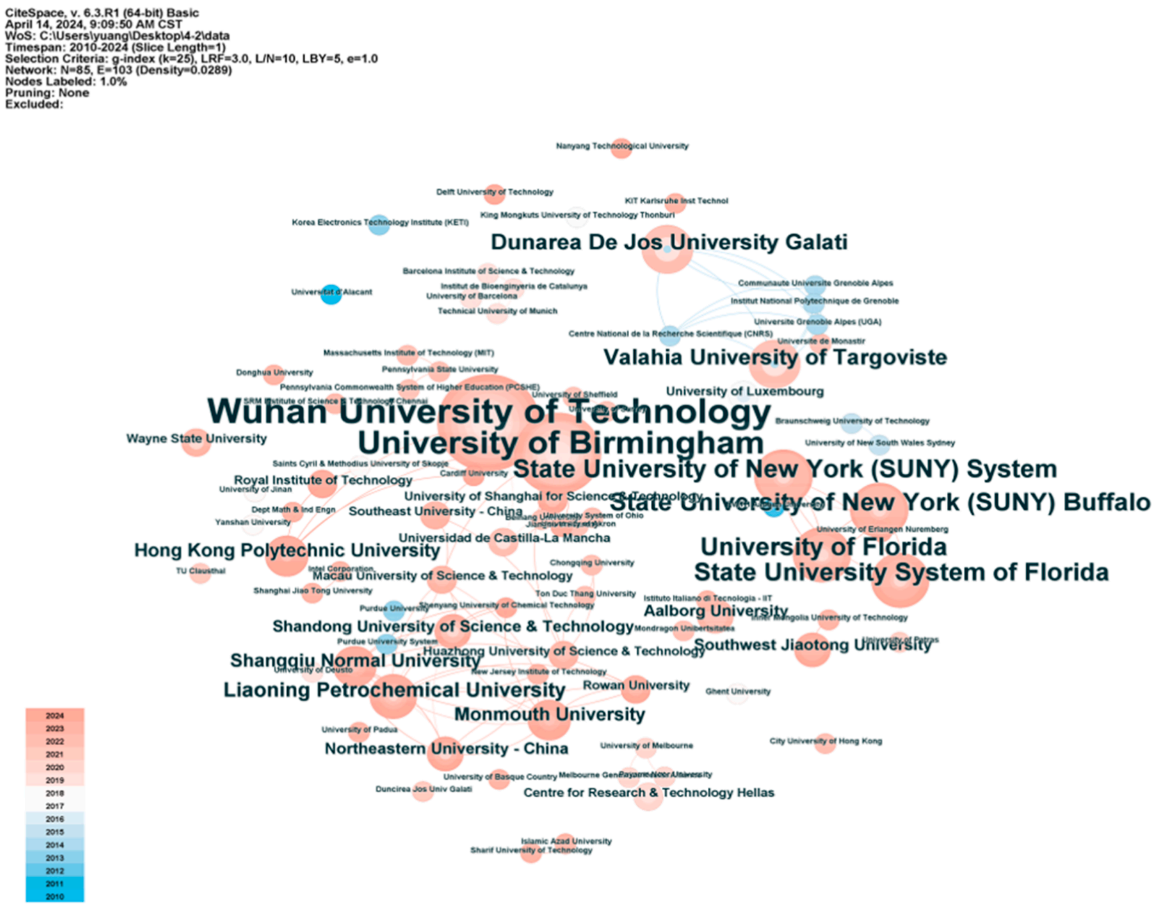

Fig. 7. Affiliations co-occurrence knowledge graph.

keywords characterised by their high frequency include human robot collaboration, design, model, and framework. The above discussion indicates that HRCD research is centred around the theoretical design of architecture, model-based systems engineering, digital engineering, etc. The focus lies in advancing the field through intricate and innovative approaches to these areas of study. It is worth noting that the pivotal keywords revolve around lithium-ion batteries, signifying the heightened focus of HRCD on recycling research within the realm of new energy. This aligns seamlessly with the overarching ideals of eco-friendliness, a reduced carbon footprint, and sustainable progression. Moreover, disassembly sequence planning and algorithms have shown increased frequency and significant co-occurrence with digital twins and artificial intelligence. This observation highlights the growing interest in the role of I5.0 technology within HRCD research.

3.3.4. Keywords co-occurrence time domain analysis

We utilised the functionality of the CiteSpace time zone chart to acquire the temporal progression of HRCD research, as depicted in Fig. 9. Diverse time zones are established in accordance with the temporal occurrence of keywords, and their placement along the temporal axis is to the right. Moreover, the keyword font size corresponds to its frequency. As illustrated in Fig. 9, a surge of HRCD research took place approximately 2010. However, the research content at that time is rather simplistic, largely attributed to the enforcement of the collaborative robot-assisted manufacturing standard in 2009. This facilitated a favourable environment for such studies. During the period from 2010 to 2015, there was a noticeable emergence of keywords such as design, robot guidance, and computer vision. This led to a gradual shift in research focus within the field of HRCD towards the development of design-based assisted robot disassembly techniques. During the period from 2016 to 2020, there was a notable emergence of specific keywords such as optimisation, artificial intelligence, cognitive robotics, genetic algorithms, and bee colony. This phenomenon highlights the heightened focus and attention that scholars have placed on the research of HRCD in algorithm-based optimisation. Since 2020, notable keywords such as digital twins, machine learning, and industry 4.0 have emerged. The domain of HRCD has embarked upon a new phase in the realm of research trends, witnessing a marked increase in the breadth and depth of keywords while sustaining a continuous surge in research interest. The integration of HRCD with advanced technologies, including digital twins, machine learning and concurrent exploration, into Industry 4.0-related interaction design has gradually begun to take shape.

4. Content analysis

The HRCD problem has been thoroughly scrutinised and categorised into six primary domains: interaction level (co-existence, cooperation, collaboration), state (static, dynamic), disassembly type (disassembly sequence planning, disassembly line balancing, disassembly evaluation, disassembly workcell), optimisation objective (time, economic, environmental, ergonomics), solution approaches (heuristics/meta-heuristic, demonstration-based learning, virtual simulation), disassembly product (case studies, random test). This work systematically elucidates and categorises each subcategory, providing a concise summary of the findings within these categories. To dissect the technical details of the studies, we selected 39 papers published in JCR Q1 journals and top conferences in recent years for a thorough analysis. Table 2 presents an in-depth analysis of the HRCD problem across 39 studies, detailing its characteristics. The subsequent sections below offer a detailed examination of the six major categories outlined in Table 2.

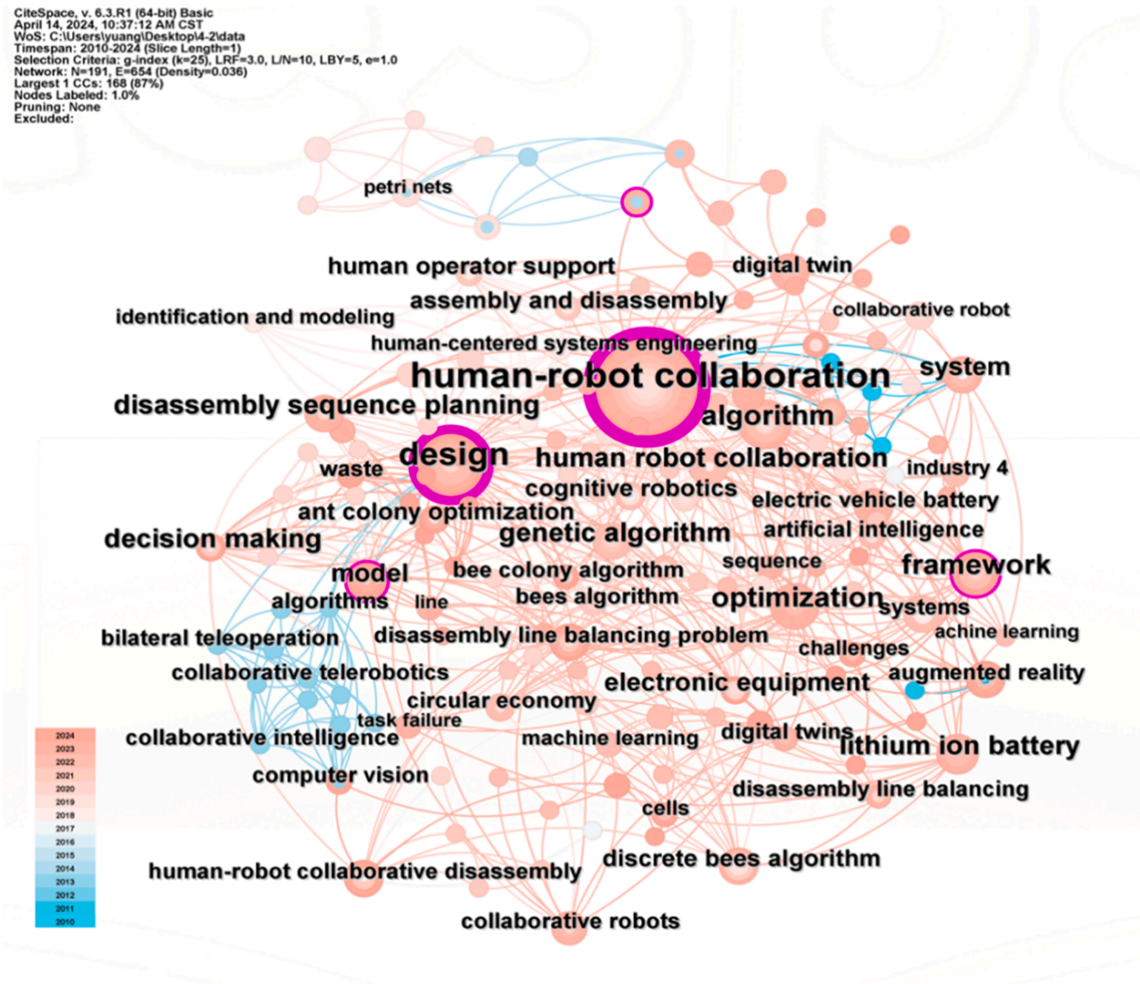

Fig. 8. Keywords co-occurrence knowledge graph.

4.1. HRCD interaction level

To mitigate ergonomic risks and increase productivity, the HRCD type is introduced in the disassembly process. Among the 39 papers, more than 80 % focused on human-robot collaboration scenarios, as illustrated in Table 2. The majority of HRCD studies have primarily offered conceptual frameworks or conducted trial case studies. However, the definitions of HRCD in studies are inconsistent and lack a universally accepted consensus. Table 3 provides an overview of the pertinent categories and detailed descriptions pertaining to HRCD. Based on the attributes of autonomy and diversity in human–robot collaboration (HRC), Parsa et al., 2021 [49] classify HRCD into four distinct categories: leadership, support, inactive, and intuitive. The relationships between leaders and followers, whether human or robotic, can manifest in various forms of collaborative interaction [75]. These entities have the ability to adapt their roles based on the specific task at hand and the overall requirements of the situation. In accordance with the operational attributes of both humans and robots, encompassing working space, task allocation, time management, and context awareness, Wu et al. [39] categorize HRCD into four distinct types: coexistence, sequential cooperation, parallel cooperation, and collaboration. According to Zhang et al. [76], the classification of HRCD types among robots, operators, and tasks encompasses four distinct categories: independence, sequential, simultaneous, and supportive. This systematisation provides a nuanced framework for understanding the complex dynamics at play within robotic operations. Villani et al. [77] provide an overview of the distinctions among safety, coexistence, and collaboration in the interplay between humans and robots. Overall, the spectrum of HRCD ranges from coexistence within physical workspaces, with minimal task sharing, to full-fledged cognitive participation in shared tasks. It is imperative to underscore the fundamental necessity of ensuring and implementing safe human conduct across all scenarios.

4.2. Disassembly state

The factors involved in the process of HRCD can be broadly categorised as static or dynamic. Static factors encompass measurable and predictable elements, such as the physical properties of products and geometric constraints [78]. On the other hand, dynamic factors are characterised by their unpredictability and potential for change, encompassing variables such as the degree of fastener failure and component demand [79,80]. The distribution of HRCD states presented in Table 3 reveals that 75 % of the studies are centred on static conditions, whereas 25 % pertained to dynamic optimisations.

The quality of EOL products is ambiguous, with inconsistent difficulty in component disassembly and varying operator skill levels [81]. Consequently, the static modelling methods proposed in most studies are unable to accommodate the uncertainty inherent in the disassembly process [82]. Several scholars have examined the intricacies involved in formulating dynamic HRCD strategies under uncertain and ever-changing conditions. Amirnia et al. [43] endeavor to enhance the optimization of real-time dynamic DSP through the introduction of a context-aware recommendation system. To delineate the dynamic nature of addressing uncertain product quality and operator capability, Jin

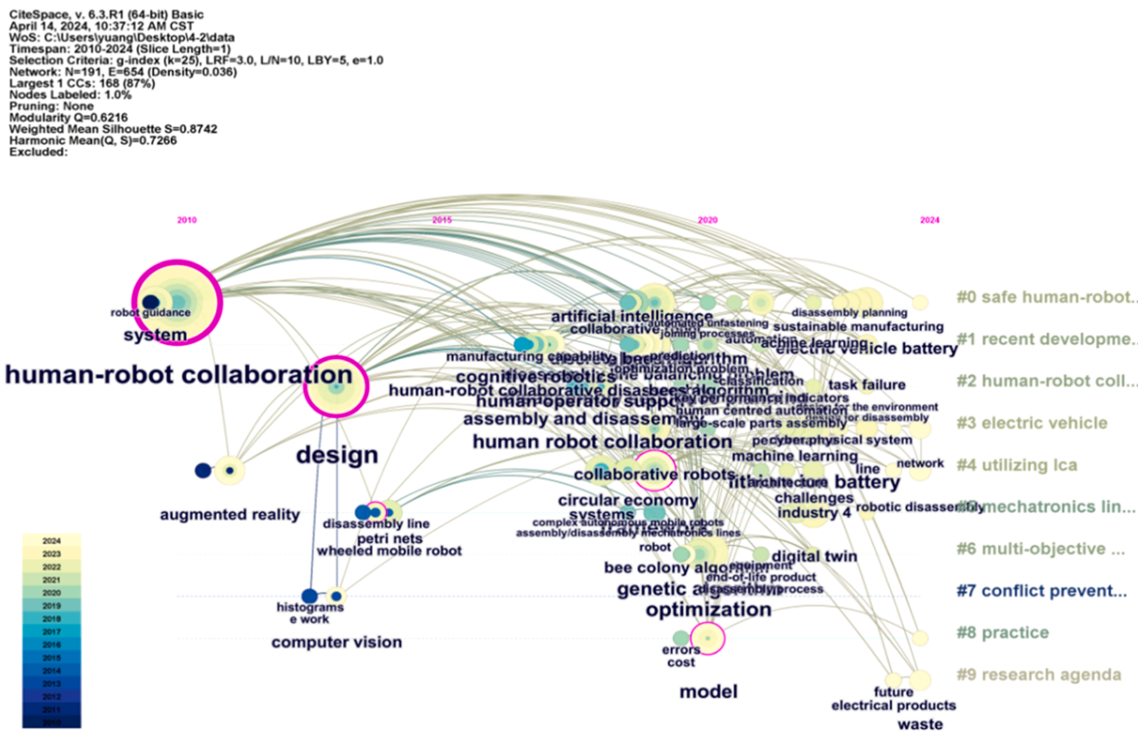

Fig. 9. Keywords co-occurrence time domain graph.

et al. [37] introduce a task-based dynamic DLB optimization model meant to monitor and adapt to changes within a fluid disassembly environment. Liao et al. [57] have put forth an optimal model for the planning of disassembly sequences, while considering the uncertainty surrounding HRCD. This innovative approach successfully achieves a harmonious balancing between disassembly cost, security, and the complexity of the disassembly process. Yuan et al. [58] present a dynamic assessment method for HRCD resilience, which incorporates stability, redundancy, efficiency and adaptive measures. Moreover, it is important for the HRCD system to consider the robot's understanding of the task, as well as its ability to work ergonomically and perform effectively in a team when executing tasks. Some scholars have studied human visual autonomous perception [72], mutual-cognitive for human and robotic [70], human motion prediction [71,73,74], and human activity recognition [69]. However, these technologies are only suitable for the experimental environment operation of simple actions, and lack the ubiquitous perception ability of complex environment and disturbance factors.

4.3. Disassembly type

The process of disassembly planning entails determining the sequential disassembly [83], process implementation, production line layout [84], and disassembly workcell design in compliance with specific requirements. The objective is to formulate a strategy that can maximize anticipated value within certain limitations. Scholars have conducted systematic research on HRCD, which is described in the following aspects: disassembly sequence planning, disassembly line balancing, disassembly evaluation and disassembly workcell design. In particular, scheduling, sequence planning, and line balancing all involve optimization problems, but with different objectives and approaches. Disassembly scheduling is to coordinate the allocation of resources and the sequencing of tasks to ensure efficient production. Disassembly sequence planning focuses on optimizing the sequence of operations to achieve the desired goal optimization. Disassembly line balancing is to ensure that the process flow and resource allocation are reasonable, so as to improve the imbalance of the production process. The combination of the three disassembly types can significantly improve the overall efficiency of the production system.

4.3.1. Disassembly sequence planning

HRCD disassembly sequence planning aims to establish the most efficient disassembly sequence by developing a detailed product disassembly information model that aligns with the requirements and limitations of individual parts [85,86]. Optimisation, in this context, pertains to the objectives pursued by decision-makers, such as minimising time [87], reducing costs [88], maximising recovery value from dismantled components [89], and mitigating environmental impact [90]. It is noteworthy that the disassembly sequence entails uncertainties that are inherently challenging to forecast [54]. Additionally, the profiling of workers' attributes proves elusive at the initial stages of the disassembly process [47]. Considering ergonomic factors, the planning of the sequence for HRCD becomes more intricate [46]. Drawing on the physical attributes and geometric limitations of EOL products, studies like Amirnia et al. [43] proposed a real-time multi-agent deep reinforcement learning model tailored to address the intricacies and unpredictability inherent in HRCD tasks. Chu et al. [44] conducted a study on the optimisation of coupling in the HRCD process, focusing on scheduling sequences, disassembly processes and policy assignments. In a similar vein, Fang et al. [52] delved into a location-constrained HRCD planning model that integrates aspects such as DSP, DLB and robot path planning. Additionally, Qu et al. [48] developed an HRCD environment that incorporates virtual and real interactions to dynamically generate real-time cooperative strategies, resulting in significantly enhanced flexibility within HRCD operations. The existing methods for disassembly sequence planning are based on qualitative operation optimization, which leads to the distortion of the optimization solution under uncertain conditions and lacks an adaptive optimization algorithm that integrates dynamic data. This limitation hinders the effectiveness of these methods in practical application scenarios.

Table 2
The summary of related works.

References		Interaction level			State		Disassembly type				Optimization objective				Solution approach			Product	
		CO	CP	CL	ST	DY	DSP	DLB	DE	DC	TM	EC	EN	ER	HC	DL	VS	CS	RT
[36]	Guo et al. 2023a	√			√			√				√	√		√				√
[37]	Jin et al. 2022		√			√		√			√			√	√				√
[38]	Wu et al. 2022	√			√			√			√	√		√	√			√	
[39]	Wu et al. 2024			√	√			√			√				√				√
[40]	Xu et al. 2021			√	√			√			√	√			√			√	
[41]	Xu et al. 2020a			√	√			√			√	√	√		√			√	
[42]	Yin et al. 2024	√			√			√			√			√	√			√	
[43]	Amirnia et al. 2023	√				√	√				√				√			√	
[44]	Chu et al. 2023				√		√				√				√			√	
[45]	Lee et al. 2022			√	√		√				√						√	√	
[46]	Lou et al. 2024			√	√		√				√			√	√			√	
[47]	Lee et al. 2024			√	√		√				√			√	√			√	
[48]	Qu et al. 2024			√	√		√							√	√			√	
[49]	Parsa et al. 2021			√	√				√		√				√			√	
[50]	Xu et al. 2020b			√	√		√				√	√			√			√	
[51]	Belhadj et al. 2022			√	√		√				√						√	√	
[52]	Fang et al. 2023			√	√		√	√			√		√	√	√				√
[53]	Lee et al. 2020			√	√		√				√				√				√
[54]	Guo et al. 2023b			√	√		√				√	√			√			√	
[55]	Li et al. 2020	√			√		√				√					√		√	
[56]	Zhou et al. 2022			√	√		√				√			√	√			√	
[57]	Liao et al. 2023			√		√	√				√			√			√	√	
[58]	Yuan et al. 2023			√		√			√		√			√	√			√	
[59]	Fan et al. 2024			√	√				√					√		√		√	
[60]	Cheng et al. 2017			√	√				√			√	√		√				√
[61]	Yuan et al. 2024			√		√	√				√	√			√			√	
[62]	Qin et al. 2024			√	√			√			√	√				√		√	
[63]	Wang et al. 2024			√	√				√		√					√		√	
[64]	Huang et al. 2021			√	√					√	√					√		√	
[65]	Ding et al. 2019			√	√					√	√			√	√			√	
[66]	Prioli et al. 2023			√	√					√					√			√	
[67]	Fan et al. 2022			√	√					√				√		√		√	
[68]	Liu et al. 2019a			√	√		√			√						√		√	
[69]	Zhang et al. 2024			√		√				√				√		√		√	
[70]	Li et al. 2024			√		√				√				√		√		√	
[71]	Lou et al. 2024		√			√	√				√	√			√				√
[72]	Deng et al. 2024b			√		√				√	√					√		√	
[73]	Guo et al. 2024	√			√			√			√			√	√			√	
[74]	Liu et al. 2023			√		√				√				√		√		√	

Interaction level: CO: Co-existence, CP: Cooperation, CL: collaboration.
Problem characteristics: ST: Static, DY: Dynamic.
Disassembly type: DSP: Disassembly sequence planning, DLB: disassembly line balancing, DE: disassembly evaluation, DC: disassembly workcell.
Optimization objective: TM: time, EC: economic, EN: environmental, ER: Ergonomics.
Solution approaches: HC: heuristics/meta-heuristic, DL: demonstration-based learning; VS: virtual simulation.
Product object: CS: case studies, RT: random test

4.3.2. Disassembly line balancing

The problem of disassembly line balancing involves the allocation of each task to its corresponding workstation to fulfil all priority relationships between tasks and optimise specific effectiveness measures [91,92]. The HRCD line balancing problem is influenced by numerous uncertainty factors, including constraints [93], component requirements [94], product conditions [95], and reliability. The majority of scholars choose to utilise preexisting frameworks or simulation models in the development of human–robot collaborative disassembly lines [42]. Furthermore, the predominant focus in most research is on safety as a primary performance criterion for assessing HRCD systems. Lee et al. [45] and Liao et al. [57] endeavoured to delegate the most perilous undertakings to auxiliary robots with the aim of enhancing the wellbeing of human labourers engaged in disassembly operations. Guo et al. [36] proposed a collaborative HRCD line balancing model that considers the unpredictable nature of task times to mitigate time uncertainty and minimise component value loss. Wu et al. [39] incorporated HRC models for various collaborative scenarios into the optimisation problem of disassembly line balancing. Xu et al. [40] conduct a study on the line balancing model of HRCD with security policies. Current research in the field of disassembly line balancing primarily focuses on a single type of collaborative robot, with limited attention given to detailed task allocation for the collaborative disassembly of multi-station hybrid robots.

4.3.3. Disassembly evaluation

Disassembly evaluation is a multi-attribute decision-making process for a given feasible scheme, which effectively helps decision makers make rational process planning, production line layout [96], and performance assessment [97]. The existing research on disassembly evaluation can be divided into two main groups: disassemblability evaluation and disassembly scheme evaluation. The evaluation of disassemblability involves both qualitative and quantitative assessments to determine the ease of disassembly for a given product. This process lays the groundwork for disassembly and serves as a crucial component of sustainable remanufacturing practices. Directly assessing the merits and demerits of Pareto dismantling schemes derived from optimised solutions is often challenging for decision makers [98]. Consequently, several academics have investigated the formulation of an evaluation index system and assessment methods for disassembly decisions. The assessment of HRCD manufacturing capability is fundamental to realising disassembly optimisation, serving as a solid foundation for workshop scheduling optimisation. For instance, Cheng et al. [60] developed an HRCD manufacturing capability evaluation model that seamlessly

Table 3
HRCD related categories and descriptions.

Reference	Classification	Description
Wu et al 2024	Coexistence	No shared workspace; unrelated task; no shared time; without contact; no context awareness
	Sequential cooperation	No shared workspace; linked task; no shared time; without contact; no context awareness
	Parallel cooperation	Shared workspace; unrelated task; no shared time; without contact; context awareness
	Collaboration	Shared workspace; shared task; shared time; contacted; context awareness
Parsa et al 2021	Leading	Both humans and robots lead the task.
	Supportive	Human or robot act as a follower to perform on the demand action.
	Inactive	The task is performed when one of the humans or robots is on standby.
	Intuitive	By being intuitive and adaptable to tasks, humans and robots can switch roles.
Zhang et al 2021	Independent Collaboration	A human operator and a cobot work on separate work pieces independently
	Sequential Collaboration	A human operator and a cobot carry out sequential tasks in serial processes on the same work piece.
	Simultaneous Collaboration	A human operator and a cobot work on the same work piece at the same time but on separate processes.
	Supportive Collaboration	A human operator and a cobot work together on the same work piece interactively
Villani et al 2018	HCR Safety	Collision avoidance, collision detection and reaction
	HCR coexistence	Robots workspace shared with humans
	HCR collaboration	Coordination of actions and intentions

integrates historical data with real-time data, representing a significant advancement in the field. Yuan et al. [58] formulated a disassembly evaluation model that integrates fuzzy Bayesian and analytical network process. This model meticulously categorises the indicators of HRCD disassembly schemes, with the crucial aspects of time, economy, and environment. To achieve the perception of human-object features during HRCD, Fan et al. [59] proposed an advanced method for intensive pose estimation through collaborative efforts between humans and robots, demonstrating a sophisticated awareness of occlusion. The current methods for evaluating disassembly schemes are only suitable for assessing and deciding the disassembly sequence within a single time segment, lacking a systematic and comprehensive evaluation method for HRCD schemes under varying time conditions.

4.3.4. Disassembly workcell

A disassembly workcell integrates multiple devices and systems, including robots, automation equipment, and information processing systems, in collaboration with human operators. It is an efficient and cooperative work system that combines human intelligence and robot skills [99], and is highly integrated, intelligent, cooperative and flexible [100]. It can significantly increase productivity, reduce error rates, optimise decision-making and risk control, and improve user experience [101]. HRCD has been extensively employed in industry, yet it remains in the experimental stage within laboratory settings. This complexity arises primarily from the challenges in accurately perceiving human intentions, robot trajectories, real-time operational dynamics and product states. Owing to the exponential advancements in artificial intelligence and computing power, there has been a burgeoning trend within HRCD towards leveraging these cutting-edge technologies to enhance operational efficiency and foster greater collaboration.

This development represents a paradigm shift in the approach to disassembly processes as businesses increasingly recognise the transformative potential of AI and advanced computing tools. Emerging technologies, such as mixed reality (MR) [102,103], augmented reality (AR) [104], digital twins (DT) [105], reinforcement learning (RL) [106], etc., provide a theoretical framework for HRCD symbiosis. Some related HRCD framework systems are present in Table 4. Liu et al. [68] presented a five-phase HRCD framework, including perception, cognition, decision, execution and evolution, reflected in two internal circles, a human-in-loop circle and a robot-in-loop circle, and one external circle, namely, PCDEE-Circle. Deng et al. [72] presented a dual-loop implementation architecture that integrates human-in-the-loop strategies and deep active learning techniques, namely, learning-by-doing HRCD. Li et al. [70] presented a system architecture of MR-enabled mutual-cognitive HRCD that integrates ergonomic, visual reasoning, and cognitive services. To address the unknown state of recognising human activity in the disassembly process, studies like Zhang et al. [69] proposed an unsupervised learning HRCD workcell for action features directly from videos. Lou et al. [46] proposed a human-cyber-physical system (HCPS) framework for HRCD by illustrating human-in-the-loop (HitL) and human-on-the-loop (HotL) paradigms. Qu et al. [48] established an HRCD dynamic adaptive planning framework that included a digital twin environment, disassembly sequence planning, and feature cross-domain recognition.

4.4. Optimisation objective

The objective functions examined in these studies can be divided into four categories: time, economy, environment, and ergonomics. A total of 15 distinct sub-objective functions were delineated across a review of 39 pertinent studies, as shown in Fig. 10. Table 5 provides an overview of the optimization objectives utilised in the HRCD survey, encompassing the objective function category, symbol, and extreme value. In this study, the time objective is further classified as total/cycle time,

Table 4
The related HRCD framework systems.

Reference	Framework	Description
Liu et al 2019	DEE-circle	It embodies the decision making, execution, and evolution, which are exactly the kernel to realise physical.
	PCDE-circle	It contains multi-modal perception, multi-target cognition, strategy and decision making and control.
	PCDEE-circle	It contains perception, cognition, decision, execution and evolution.
Deng et al 2024b	Learning-by-doing circle	It is realised by a dual-loop implementation architecture, where the doing loop denotes the human-robot collaborative disassembly loop and the learning loop denotes the deep active learning loop.
Li et al 2024	MR-enabled mutual-cognitive	It is consisted of ergonomic collaboration in physical spaces, visual reasoning modules and virtual replicas in cyber spaces, and cognitive services in MR spaces.
Zhang et al 2024	Unsupervised learning HRCD	It combines the ability of CNNs to parse spatial image data and recurrent neural networks (RNNs) to handle temporal data.
Lou et al 2024	HCPS-HitL-HotL	The HitL paradigm focuses on operators in HRCD, such as interacting with the robot, disassembly situation awareness, and emergency handling, etc. The HotL paradigm values expert decision making and domain knowledge when dealing with frequent disassembly changes, such as high-level decision making, dealing with uncertain information, knowledge-based reasoning, etc.
Qu et al 2023	DT-HRCD adaptive planning system	Virtual space provides digital modeling and simulation, whereas physical space can receive control instructions from virtual space in real time and modify the behavior of disassembly.

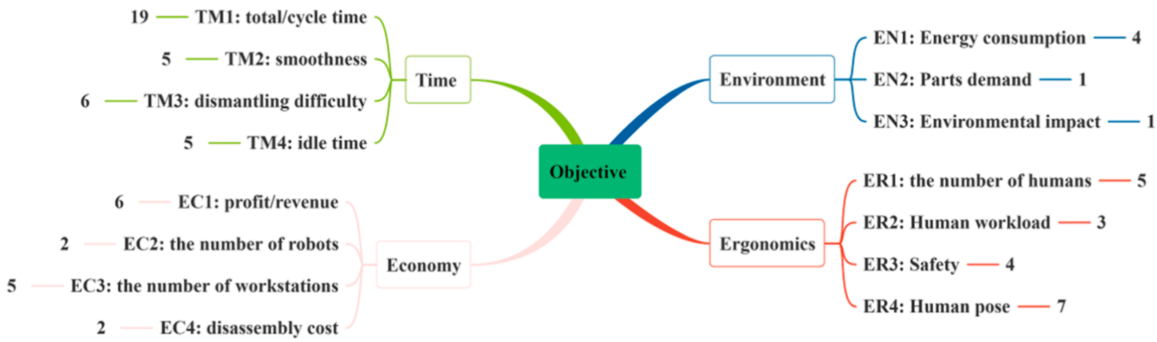

Fig. 10. Optimisation objective categories and specific instances.

Table 5
Exhaustive analysis of the optimization objectives.

Publications	Time				Economic				Environmental			Ergonomics			
	TM1	TM2	TM3	TM4	EC1	EC2	EC3	EC4	EN1	EN2	EN3	ER1	ER2	ER3	ER4
Max(+)/Min(-)	-	-	-	-	+	-	-	-	-	+	-	-	-	+	+
Guo et al. 2023a					√				√						
Jin et al. 2022	√					√						√			
Wu et al. 2022		√			√		√					√			
Wu et al. 2024		√					√	√							
Xu et al. 2021		√					√			√					
Xu et al. 2020a			√		√		√		√						
Yin et al. 2024							√					√			
Amirnia et al. 2023			√												
Chu et al. 2023	√														
Lee et al. 2022	√														
Lou et al. 2024	√		√		√								√		
Lee et al. 2023b						√						√			
Qu et al. 2023				√											
Parsa et al. 2021	√														
Xu et al. 2020b	√		√					√							
Belhadj et al. 2022	√														
Fang et al. 2023	√								√			√			
Lee et al. 2020	√		√												
Guo et al. 2023b	√				√										
Li et al. 2020	√														
Zhou et al. 2022	√													√	
Liao et al. 2023				√											
Yuan et al. 2023	√	√												√	
Fan et al. 2024															√
Cheng et al. 2017	√	√	√						√		√			√	
Yuan et al 2024	√	√											√		
Qin et al 2024					√										
Wang et al 2024		√													√
Huang et al. 2021	√			√											
Ding et al. 2019	√			√									√		
Fan et al. 2022													√		
Liu et al. 2019a															√
Zhang et al. 2024															√
Li et al. 2024														√	√
Lou et al 2024	√				√										
Deng et al. 2024	√														
Guo et al 2024		√													
Liu et al. 2023															√

TM1: total/cycle time; TM2: smoothness; TM3: dismantling difficulty; TM4: idle time
EC1: profit/revenue; EC2: the number of robots; EC3: the number of workstations; EC4: disassembly cost
EN1: Energy consumption; EN2: Parts demand; EN3: Environmental impact
ER1: the number of humans; ER2: Human workload; ER3: Safety; ER4: Human pose

smoothness, disassembly difficulty, and idle time. The functions of economic goals can be divided into categories: profit/revenue, the number of robots, the number of workstations, and disassembly cost. The features related to environmental goals include energy consumption, part demand and environmental impact. The features related to ergonomic goals include the number of humans, human workload, safety, and human pose. The exhaustive analysis of the objective function, depicted in Table 5, primarily focuses on the temporal and human-centric objectives. Approximately 66 % pertains to time-related goals, 23 % to economic considerations, 10 % to environmental factors, and 46 % to human-centric elements. HRCD ensures that the working environment is designed to be ergonomically sound and prioritises the safety and well-being of all employees, thereby reducing time waste, minimising costs, and minimising environmental impact.

4.5. Solution approaches

This section categorises and analyses the solutions to the HRCD problem. HRCD solutions fall into three main categories (as Table 2): heuristics/meta-heuristic (62 %), demonstration-based learning (28 %), and virtual simulation (10 %). Among the three methods, heuristic and metaheuristic algorithms, such as genetic algorithms, particle swarm algorithms, and bees algorithms, stand out as the primary approaches for addressing HRCD. Nevertheless, intelligent algorithms often exhibit suboptimal local search capabilities or a lack of global search abilities. Therefore, it is imperative to amalgamate the problem-solving abilities of multiple intelligent algorithms to effectively address large-scale problem-solving tasks. The current heuristic solution methods are predicated upon operation optimisation within qualitative parameters, ultimately leading to distortion in optimisation solutions under uncertain conditions and a deficiency in adaptive optimisation through dynamic data integration.

The virtual simulation method entails constructing a disassembly environment for physical entities and their product characteristics. This allows for the realisation of disassembly simulations within virtual environments and under constraint conditions, ultimately leading to the identification of optimal decisions based on simulation. Virtual simulations offer operators the opportunity for trial and error to uncover potential ergonomic hazards. However, HRCD is a complex system with dynamic factors, making it challenging to accurately replicate the real disassembly state through virtual simulation methods. Moreover, the presence of uncertainty in human factors amidst time-varying conditions adds a layer of complexity to the construction of the HRCD model, rendering it a more challenging and intricate task. The demonstration-based learning method is a sophisticated process of acquiring knowledge that aims to facilitate human operators in instructing robots how to adeptly perform complex tasks, cultivate autonomous visual perception in robots, and enhance the interaction between humans and robots. In addition, emerging technologies, such as mixed reality, augmented reality, and computer vision, provide solution for the symbiotic relationships of the human and robot [70]. The training of learning models necessitates the utilisation of diverse and abundant datasets to effectively encapsulate their characteristics. Owing to the significant level of uncertainty and intricacy associated with EOL products, the dearth of labelling data and the inefficaciousness of data collection are pivotal impediments to learning methodologies. In addition, it is imperative that either the robotic entity or the human agent adhere to predefined directives throughout the task execution.

4.6. Disassembly product

In research studies, experiments are conducted using test data to validate the accuracy and effectiveness of the proposed solution methods, to perform sensitivity analysis, and to demonstrate the applicability of the proposed approach by comparing it with other solution methods. This work examines test data types, which are classified into two categories: case studies and random test problems. Among the references on the HRCD problem, 82 % exclusively considered case studies, and 18 % solely employed random test problems. The most prevalent form of test data found in the academic literature is case studies.

The distribution of products utilised in addressing the HRCD problem is delineated in Table 6. Xu et al. [40] proposed a human-robot collaborative disassembly line balancing model considering safety policy, and used discrete bee algorithm to realize bearing coupler sequence planning. Chen et al. [44] used Q-learning algorithm to optimize the human-robot task allocation strategy in the disassembly process of lithium battery pack. Lou et al. [46] proposed a human-cyber-physical system framework for a human-robot collaboration cell, and a control box is presented to illustrate the feasibility of the proposed approach. Yuan et al. [58] proposed a multi-criteria resilience assessment method of HRCD for supporting spent lithium-ion battery recycling. Existing case studies have not sufficiently considered the accuracy of task execution in complex environment disassembly. The experimental environment can be expanded to include more disassembly workshops in order to accommodate the execution of tasks in complex scenes.

Table 6
The distribution of cases utilised in addressing the HRCD problem.

Reference	Product	Solution algorithm
Wu et al. 2022	Tesla Model S battery	NSGA-II
Xu et al. 2021	Bearing coupler	The improved discrete bees algorithm
Xu et al. 2020b	Hammer drill	Artificial bee colony algorithm
Yin et al. 2024	Refrigerators, microwave ovens, and dishwashers	Problem-oriented group evolutionary algorithm
Amirnia et al. 2023	Hard disk drive	Deep reinforcement learning
Chu et al. 2023	Tesla Model S battery	Hybrid particle swarm optimization with the Q-learning algorithm
Lee et al. 2022	Hard disk drive	Comprehensive disassembly sequence planning algorithm
Lou et al. 2024	Control box of AMR	Improved hybrid grey wolf optimization algorithm
Lee et al. 2023b	Toybox, hard disk drive	Disassembly task planning algorithm
Qu et al. 2023	Lithium-ion batteries	Genetic algorithm
Parsa et al. 2021	Fuel pump	PROMETHEE II method
Xu et al. 2020a	Simplified computer	Modified discrete bees algorithm based on Pareto
Belhadj et al. 2022	Gear box	Sequential approach
Guo et al. 2023b	Hammer drill, copying machine, and radio	Shuffled frog leaping algorithm
Li et al. 2020	Turbocharger	Spiral search strategy
Zhou et al. 2022	Removing screws	Particle swarm optimisation-Pareto algorithm
Liao et al. 2023	Dell desktop computer	Utility function
Yuan et al. 2023	Chevrolet Bolt battery	Fuzzy Bayesian
Yuan et al. 2024	Simplified satellite model	Artificial bee colony algorithm
Qin et al. 2024	Lithium-ion batteries	Mixed integer programming
Wang et al. 2024	Lithium-ion batteries	Convolutional neural network
Huang et al. 2021	Turbocharger	Experimental
Ding et al. 2019	Roller chain	Graph-based knowledge
Prioli et al. 2023	LED box	Experimental
Fan et al. 2022	Lithium-ion batteries	Multi-granularity scene segmentation network
Liu et al. 2019a	Diaphragm coupling	Multi-modal perception, multi-target cognition
Zhang et al. 2024	Hard disk drive	Sequential variational autoencoder, human action recognition
Li et al. 2024	Lithium-ion batteries	MR, visual reasoning
Deng et al. 2024b	Lithium-ion batteries	Augmented reality (AR) glasses
Guo et al. 2024	Engine	Grey wolf optimization algorithm
Liu et al. 2023	Hard disk drive	Task-constrained motion planning

The following compilation presents a selection of the most sought-after products, listed in descending order of their prevalence within the current literature: lithium-ion batteries [48,63,72], hard disk drive [68,69], computer [53,74], turbocharger [64]. On the other hand, all products can be categorised into four distinct classes, as Fig. 11, lithium-ion batteries (LIB) (with 29 %), fuel vehicle parts (FVP) (29 %), waste electrical and electronic equipment (WEEE) (32 %), and others

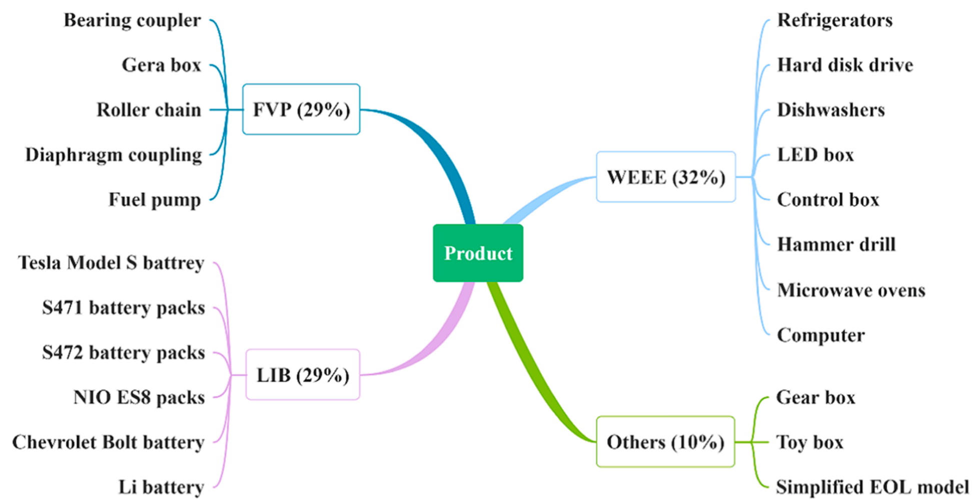

Fig. 11. Disassembled product categories and specific instances.

(10 %). Products categorized as "other" have only been documented in a limited number of studies, including toyboxes and simplified EOL model. In general, the current HRCD research products are mainly concentrated in fuel vehicle parts, WEEE, and lithium-ion batteries. The disassembly of lithium-ion batteries has garnered increasing attention, a phenomenon closely intertwined with the rapid advancement of electric vehicles and the maturation of battery technology.

5. Discussion

5.1. Discussion around RQ1

Through in-depth analysis of literature content and visual presentation of measurement results, we can clearly obtain the primary themes and presentations in HRCD. The findings of the literature contents analysis can be encapsulated as follows.

(1) Static disassembly environments have prevailed in the realm of research, although there is undeniably an inclination towards dynamic disassembly states. With dynamic disassembly, the uncertainty factor that garners the most attention is variability in product quality, closely followed by uncertainties stemming from ergonomics. HRCD extends from initial cooperation to symbiotic operations, where operators transition from simple cooperative disassembly to immersive disassembly operations. This shift realizes the replacement of an economy-dominated production mode with a focus on human comfort. With the increasing requirements of maintainable disassembly and remanufacturing, the disassembly process should not only meet the operational experimental conditions, but also be suitable for field operations in complex social environments.

(2) DSP and disassembly workcell are two of the most researched topics, while DLB and disassembly evaluation follow suit in terms of scholarly attention. Optimisation objectives encompass various factors, such as time and human elements. Disassembly planning finalizes the allocation of qualitative operations, resulting in a set of process steps for both human and robot operations. The majority of disassembly line balancing is focused on optimizing U-shaped or linear layouts for single-type products. There is a lack of studies on evaluation methods for disassembly schemes, and the assessment and decision-making of heuristic algorithm solutions are biased. Among these, the primary objective functions are total/cycle time, profit/revenue, human pose, and the number of human operators. The initial research focuses more on the objective of disassembly time and economic benefits, and gradually transitions to low-carbon indicators such as energy consumption and environment. The proposal of carbon peak policy has produced a wave of research in pursuit of low carbon and green.

(3) The most widely adopted technology to solving HRCD problems is heuristics/meta-heuristic. Genetic algorithms, bees algorithms and particle swarm optimisation algorithms are commonly used as meta-heuristic algorithms within the realm of optimisation techniques. Although demonstration-based learning has also been advocated as a viable solution in numerous studies, their utilisation has not been as widespread as that of iterative heuristics. Furthermore, virtual simulation serves as a prevalent solution for the strategic planning of disassembly processes. With the promotion of digital twin, blockchain and Internet of things technology, artificial intelligence algorithm plays a potential mainstream trend in HRCD solution. In regard to disassembling the requisite products for an experiment, the preferred type is case studies. The disassembled products are primarily classified into three distinct categories: LIB, FVP and WEEE. Among these, lithium-ion batteries, computer and hard disk drives and turbochargers have emerged as frequently encountered materials. In particular, in the past three years, the related research on lithium battery disassembly has increased dramatically, which is closely related to the maturity and application of new energy battery technology.

5.2. Discussion around RQ2

Despite the extensive discussions and research advancements in HRCD, it is important to acknowledge its existing limitations. The existing HRC technology is mature and can provide technical support for HRCD. However, the problems of compatibility, communication delay and data missing will restrict the direct application of HRC technology in the field of disassembly. This section outlines a couple of challenges that deserve attention to realize the high precision, deep coupling, and optimal efficiency of HRCD.

(1) The reusability of the HRCD planning model and knowledge is challenging. Current HRCD design methods only consider the depiction and alteration of a solitary and unchanging relationship and not the reuse of disassembly plans for similar products. This deficiency results in inefficiency and rigidity in regard to disassembly planning for diverse products under uncertain conditions. The disassembly line incurs high start-stop costs and has a lengthy production line layout change cycle, making it unsuitable for process design involving frequent product replacements. For the mixed-flow disassembly or replacement of production lines for various types of products, the convenient invocation of the design module for product process reconstruction is a technical challenge faced by designers. To achieve the reusability of product disassembly design methods and models, and realize efficient and high-quality dynamic disassembly modeling process, there is lack of universal knowledge base and method library to meet the needs of rapid

design.

(2) The decoupling programming paradigm is incongruent with the dynamic interlinking of time-varying data. The existing research on HRCD planning has focused predominantly on the analysis of disassembly mechanism models or data models. The results of the HRCD theoretical model analysis and the algorithm solution often deviate from the actual workshop situation, resulting in the failure of the scheme based on the model and data. This narrow focus results in a fragmented coordination of data information throughout the process of disassembly modelling, sequence optimisation, and program evaluation. Ultimately, this lack of holistic consideration leads to poor consistency between disassembly planning and program execution. The data generated from the entire disassembly process serves as the foundation for guiding process improvement. The analysis of disassembly data at the interception stage will lead to potential loss of associated value and barriers to process cohesion. To enhance the usability and precision of the model and algorithm, it is imperative to delve into the multi-dimensional, multi-level dynamic mechanism, as well as the fusion method for multi-heterogeneous twin data.

(3) The dynamic optimisation and human factor assessment of HRCD lack precision and accuracy. Current research on the optimisation and assessment of HRCD focuses only on analysing the disassembly process under ideal operational conditions within a single time segment. However, this approach fails to adequately capture the adaptability of disassembly solutions in the face of unpredictable disruptions or human fatigue. There are variations in collaborative efficiency among different workers and collaborative robots, including factors such as age, gender, and learning ability. Evaluating the suitability of different roles for HRCD schemes poses challenges. In addition, the overload of disassembly operations will lead to the degradation of operation quality and may face intermittent disturbances such as component failure and safety hazards. It is imperative to explore optimisation strategies for evaluating the resilience of systems operating under continuous time-varying conditions. This endeavor aims to enhance disassembly efficiency, ensure operational safety, and mitigate system vulnerability.

5.3. Discussion around RQ3

After analyzing the content of the literature and exploring existing research challenges, we have formulated an implementable plan for future research (as Fig. 12). In addition, existing HRC on assembly and reassembly, such as cobots sensitivity control [107], multi-modal symbiotic HRC [108], human-centered manufacturing capabilities [109,110], perceptual complexity evaluation [111], can expand the application of potential HRC techniques in the field of disassembly. These measures can be used as a guide for future research and help scholars to determine the exact direction for follow-up research.

(1) Employing HRCD for multi-scale, multi-level, and multi-temporal immersive accurate dynamic modelling. In the long term, it is imperative to conduct a comprehensive study on the multi-dimensional, fine-grained model construction method of HRCD elements. Furthermore, it is essential to unveil the fusion mechanism of multi-scale, multi-level, and multi-period models with multivariate heterogeneous data in future. This will shed light on the intricate interplay between diverse mechanisms and data across different scales and time frames. In the forthcoming era, I5.0 technology, wearable sensing devices [112], and even brain-computer interfaces can be seamlessly integrated into HRCD systems to facilitate immersive disassembly environments [113]. Explicitly, the human-robot collaboration integration will encompass virtual interfaces for augmented reality, recognition of worker behavior [114], estimation of human posture[115], and seamless sharing of information flows [116]. For high-precision multi-pose disassembly tasks,

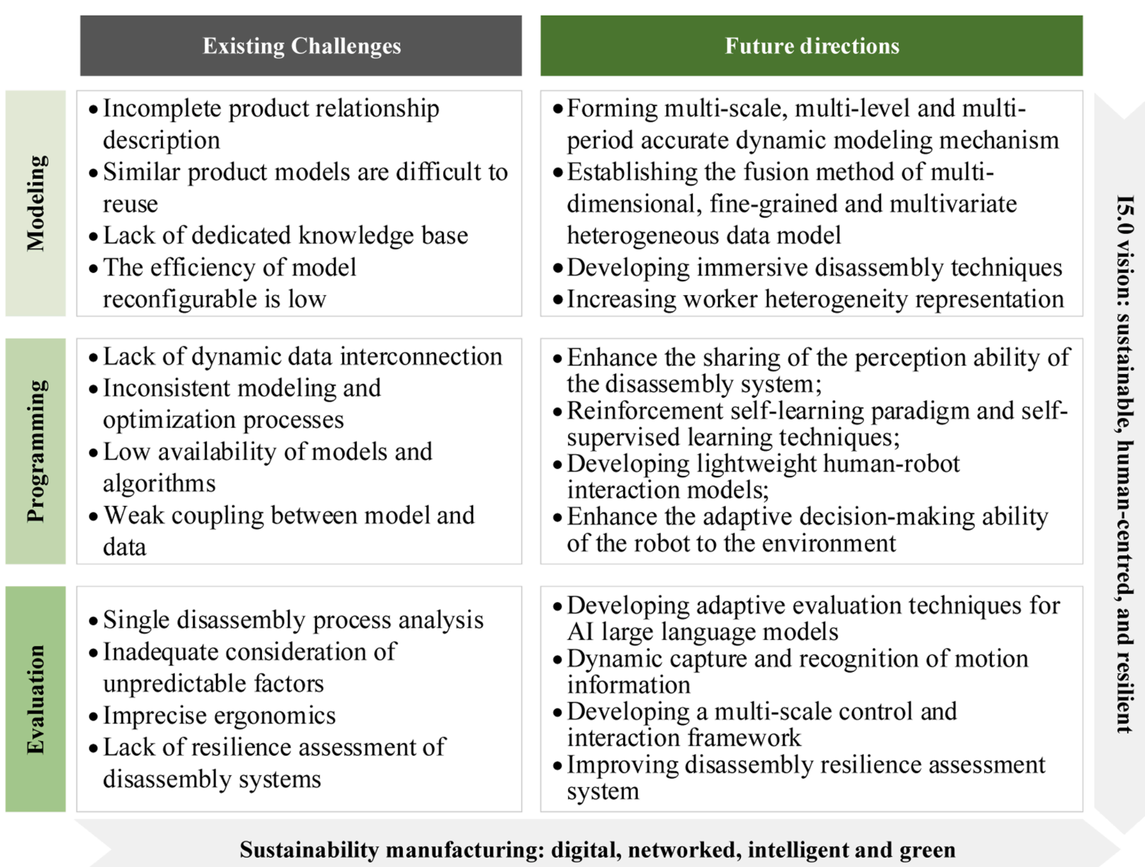

Fig. 12. Existing challenges in HRCD along with recommendations.

graph convolutional networks can improve the recognition accuracy in the HRCD topological space, such as wrist motion, wrist strength, and finger motion [117]. Finally, A more detailed representation of worker heterogeneity at multiple levels, such as age, gender, and learning ability, enables a comprehensive understanding of HRC satisfaction, productivity, and well-being [118]. By creating a sophisticated real-time mapping of the dynamic features of HRCD, the enhancement of human comfort and efficiency within the immersive HRCD environment can be achieved.

(2) Enhancing the self-supervised learning capability of task recognition and human factor perception within HRCD. The HRCD system should enhance its capacity for sharing and integrating perceptual abilities, ultimately elevating the level of cognition from perception. This will create a more sophisticated and seamless transition from perceiving information to understanding and processing it. By leveraging context awareness, visual reasoning methodologies, self-supervised learning techniques, reinforcement learning paradigms, etc., scholars are empowered to concentrate on constructing a perceptual system that ensures the safety [119], foresight, and adherence to ergonomic principles in robot operations [120]. In the future, it is possible to enhance human perception of the environment and improve our ability to meet human-centered needs, such as interpreting arm positions [121], perceiving hand-object interactions [122], learning robot trajectories in real-time, self-monitoring task execution, and capturing unmarked video data. In addition, future work can develop a lightweight human-robot interaction model to adapt to the uncertainty of the disassembly environment and the influence of external disturbances in complex scenes.

(3) Crafting methods for optimising HRCD and evaluating resilience based on the ever-changing nature of time-varying data. Future research can focus on exploring advanced optimisation and adaptive assessment techniques for HRCD, driven by AI large models [123]. By integrating digital twins [124], blockchain technology [125], and sustainable connectivity, it has become possible to evaluate the system resilience and trustworthiness of HRCD in a comprehensive manner. Specifically, based on the dynamic capture of human and robot current motion information, dynamic role assignment algorithms can be used to deal with unforeseen human behaviors [126]. Furthermore, by integrating dynamic action recognition with continuous learning technology, the HRCD framework for large-scale multi-scale control and action recognition can be further developed in the future to enhance the robustness and scalability of HRC. Moreover, it is possible to explore robot compliance control and impedance control based on physical information in order to achieve safe interaction between humans and robots.

6. Conclusion

This paper has surveyed the literature on HRCD from 2010 to 2024, employing a systematic literature review and bibliometric analysis methodology. A total of 99 final selected documents were identified through the utilisation of three popular databases. While the review primarily focuses on disassembly sequence planning, disassembly line balancing, and disassembly evaluation are included to enrich the study. The analysis of the HRCD problem is conducted based on six key categories: interaction level, state, disassembly type, disassembly objectives, solution approaches, and disassembly products. Bibliometric analysis has allowed a detailed and methodical examination of HRCD research progress, literature information mining, literature theme co-occurrence and research evolution trends.

The survey of the literature indicates that HRCD research involves multidisciplinary and interdisciplinary collaboration; thus, a network was initially established with China and the UK at its nexus. Furthermore, the core authors are widely dispersed, and several research teams dedicated to HRCD have been established. HRCD has garnered considerable attention and involvement, boasting an extensive network of research institutions, a multitude of highly regarded teams, and robust cooperative relationships. Moreover, the future trajectory of development is delineated along three primary avenues: immersive and precise dynamic modelling, temporally varying data interplay, and the perceptual and cognitive capacities of cobots. On the whole, research results will increase the in-depth understanding of smart remanufacturing and circular economy among government, academia, industry and professionals, accelerate future innovation in the green transformation of sustainability manufacturing and enhance the competitiveness of global circular value chains. In addition, despite the extensive nature of this literature review, this work is not without its research limitations. These limitations primarily arise from the restricted availability of information within certain categories of studies under examination, resulting in variations in the thoroughness and precision of categorised data during content analysis.

CRediT authorship contribution statement

Gang Yuan: Writing – original draft, Visualization, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xiaojun Liu:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Xiaoli Qiu:** Resources, Methodology, Investigation, Data curation. **Pai Zheng:** Writing – review & editing, Validation, Formal analysis. **Duc Truong Pham:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Ming Su:** Validation, Software, Resources, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing for financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Xu X, Lu Y, Vogel HB, Wang L. Industry 4.0 and Industry 5.0-inception, conception and perception. J Manuf Syst 2021;61:530–5.
[2] Leng J, Sha W, Wang B, Zheng P, et al. Industry 5.0: prospect and retrospect. J Manuf Syst 2022;65:279–95.
[3] Maddikunta PKR, Pham QV, Prabadevi B, et al. Industry 5.0: a survey on enabling technologies and potential applications. J Ind Inf Integr 2022;26:100257.
[4] Ivanov D. The Industry 5.0 framework: viability-based integration of the resilience, sustainability, and human-centricity perspectives. Int J Prod Res 2023;61(5):1683–95.
[5] Huang SH, Wang BC, Li XY, et al. Industry 5.0 and society 5.0-comparison, complementation and co-evolution. J Manuf Syst 2022;64:424–8.
[6] Leng JW, Zhong YW, Lin Z, et al. Towards resilience in industry 5.0: a decentralized autonomous manufacturing paradigm. J Manuf Syst 2023;71:95–114.
[7] Chi HR, Wu CK, Huang NF, Tsang KF, Radwan A. A survey of network automation for industrial internet-of-things toward industry 5.0. IEEE Trans Ind Inform 2023;19(2):2065–77.
[8] Destouet C, Tlahig H, Bettayeb B, Mazari B. Flexible job shop scheduling problem under Industry 5.0: a survey on human reintegration, environmental consideration and resilience improvement. J Manuf Syst 2023;67:155–73.
[9] Adel A. Future of industry 5.0 in society: human-centric solutions, challenges and prospective research areas. J Cloud Comput-Adv Syst Appl 2022;11(40):1–15.
[10] Karmaker CL, Mainul Bari ABM, Anam MZ, et al. Industry 5.0 challenges for post-pandemic supply chain sustainability in an emerging economy. Int J Prod Econ 2023;258:108806.
[11] Lu YQ, Zheng H, Chand S, et al. Outlook on human-centric manufacturing towards industry 5.0. J Manuf Syst 2022;62:612–27.
[12] Meng K, Cao Y, Peng X, et al. Smart recovery decision-making for end-of-life products in the context of ubiquitous information and computational intelligence. J Clean Prod 2020;272:122804.

[13] Tian G, Yuan G, Aleksandrov A, Zhang T, et al. Recycling of spent lithium-ion batteries: key challenges and future trends. Sustain Energy Technol Assess 2022;53:447–57.

[14] Yuan G, Yang Y, Tian G, Zhuang Q. Comprehensive evaluation of disassembly performance based on the ultimate cross-efficiency and extension-gray correlation degree. J Clean Prod 2020;245:1–13.

[15] Tian G, Zhang C, Fathollahi-Fard AM, et al. An enhanced social engineering optimizer for solving an energy-efficient disassembly line balancing problem based on bucket brigades and cloud theory. IEEE Trans Ind Inform 2023;19(5):7148–59.

[16] Yin T, Zhang Z, Wu T, Zeng Y, Zhang Y, Liu J. Multimanned disassembly line balancing optimization considering walking workers and task evaluation indicators. J Manuf Syst 2024;72:263–86.

[17] Feng Y, Cui K, Hong Z, Li Z, Yan W, Tan J. Disassembly sequence planning of product structure with an improved QICA considering expert consensus for remanufacturing. IEEE Trans Ind Inform 2023;19(5):7201–13.

[18] Zhang X, Yi D, Behdad S, Saxena S. Unsupervised human activity recognition learning for disassembly tasks. IEEE Trans Ind Inform 2024;20(1):785–94.

[19] Qu M, Wang Y, Pham DT. Robotic disassembly task training and skill transfer using reinforcement learning. IEEE Trans Ind Inform 2023;19(11):10934–43.

[20] Wang K, Li Y, Guo J, Gao L, Li X. Dynamic balancing of U-shaped robotic disassembly lines using an effective deep reinforcement learning approach. IEEE Trans Ind Inform 2024;20(4):6855–65.

[21] Liu J, Xu Z, Xiong H, et al. Digital twin-driven robotic disassembly sequence dynamic planning under uncertain missing condition. IEEE Trans Ind Inform 2023;19(12):11846–55.

[22] Hartono N, Ramírez FJ, Pham DT. Optimisation of robotic disassembly plans using the bees algorithm. Robot Comput-Integr Manuf 2022;78:102411.

[23] Lee ML, Liu W, Behdad S, et al. Robot-assisted disassembly sequence planning with real-time human motion prediction. IEEE Trans Syst Man Cybern-Syst 2023; 53. 538–450.

[24] Yin T, Zhang Z, Liang W, Zeng Y, Zhang Y. Multi-man-robot disassembly line balancing optimization by mixed-integer programming and problem-oriented group evolutionary algorithm. IEEE Trans Syst, Man, Cyber: Syst 2024;54(3):1363–75.

[25] Deng W, Liu Q, Pham DT, et al. Predictive exposure control for vision-based robotic disassembly using deep learning and predictive learning. Robot Comput-Integr Manuf 2024;85:102619.

[26] Huang J, Pham DT, Wang Y, et al. A case study in human–robot collaboration in the disassembly of press-fitted components. Proc Inst Mech Eng, Part B: J Eng Manuf 2020;234(3):654–64.

[27] Hjorth S, Chrysostomou D. Human-robot collaboration in industrial environments: a literature review on non-destructive disassembly*. Robot Comput-Integr Manuf 2022;73:102208.

[28] Ji T, Huang H, Xu X, et al. Digital twin technology-A bibliometric study of top research articles based on local citation score. J Manuf Syst 2022;64:390–408.

[29] Baroroh DK, Chu CH, Wang L. Systematic literature review on augmented reality in smart manufacturing: collaboration between human and computational intelligence. J Manuf Syst 2021;61:696–711.

[30] Liu T, Zheng P, Bao J. Deep learning-based welding image recognition: a comprehensive review. J Manuf Syst 2023;68:601–25.

[31] Liu S, Bao J, Zheng P. A review of digital twin-driven machining: from digitization to intellectualization. J Manuf Syst 2023;67:361–78.

[32] Zhong B, Wua H, Ding L, et al. Mapping computer vision research in construction: developments, knowledge gaps and implications for research. Autom Constr 2019;107:102919.

[33] Catumba BD, Sales MB, Borges PT, et al. Sustainability and challenges in hydrogen production: an advanced bibliometric analysis. Int J Hydrog Energy 2023;48(22):7975–92.

[34] Xia X, Zheng P, Li X, Gao RX, Wang L. Toward cognitive predictive maintenance: a survey of graph-based approaches. J Manuf Syst 2022;64:107–20.

[35] Giorgio A, Cola G, Wang LH. Systematic review of class imbalance problems in manufacturing. J Manuf Syst 2023;71:620–44.

[36] Guo L, Zhang Z, Zhang X. Human-robot collaborative partial destruction disassembly sequence planning method for end-of-life product driven by multi-failures. Adv Eng Inform 2023;55:101821.

[37] Jin L, Zhang X, Fang Y, Pham DT. Transfer learning-assisted evolutionary dynamic optimisation for dynamic human-robot collaborative disassembly line balancing. Appl Sci 2022;12(21):11008.

[38] Wu T, Zhang Z, Yin T, Zhang Y. Multi-objective optimisation for cell-level disassembly of waste power battery modules in human-machine hybrid mode. Waste Manag 2022;144:513–26.

[39] Wu T, Zhang Z, Zeng Y, Zhang Y. Mixed-integer programming model and hybrid local search genetic algorithm for human-robot collaborative disassembly line balancing problem. Int J Prod Res 2024;62(5):1758–82.

[40] Xu W, Cui J, Liu B, et al. Human-robot collaborative disassembly line balancing considering the safe strategy in remanufacturing. J Clean Prod 2021;324:129158.

[41] Xu C, Wei H, Guo XW, et al. Human-robot collaboration multi-objective disassembly line balancing subject to task failure via multi-objective artificial bee colony algorithm. IFAC Pap 2020;53:1–6.

[42] Yin T, Zhang Z, Liang W, et al. Multi-man–robot disassembly line balancing optimization by mixed-integer programming and problem-oriented group evolutionary algorithm. IEEE Trans Syst, Man, Cyber: Syst 2024;54(3):1363–75.

[43] Amirnia A, Keivanpour S. A context-aware real-time human-robot collaborating reinforcement learning-based disassembly planning model under uncertainty. Int J Prod Res 2023:1–22.

[44] Chu M, Chen W. Human-robot collaboration disassembly planning for end-of-life power batteries. J Manuf Syst 2023;69:271–91.

[45] Lee ML, Behdad S, Liang X, Zheng M. Task allocation and planning for product disassembly with human-robot collaboration. Robot Comput-Integr Manuf 2022;76:102306.

[46] Lou S, Zhang Y, Tan R, Lv C. A human-cyber-physical system enabled sequential disassembly planning approach for a human-robot collaboration cell in Industry 5.0. Robot Comput-Integr Manuf 2024;87:102706.

[47] Lee M, Liang X, Hu B, et al. A review of prospects and opportunities in disassembly with human–robot collaboration. J Manuf Sci Eng Trans ASME 2024;146(2):020902.

[48] Qu W, Li J, Zhang R, et al. Adaptive planning of human-robot collaborative disassembly for end-of-life lithium-ion batteries based on digital twin. J Intell Manuf 2024;35:2021–43.

[49] Parsa S, Saadat M. Human-robot collaboration disassembly planning for end-of-life product disassembly process. Robot Comput-Integr Manuf 2021;71:102170.

[50] Xu W, Tang Q, Liu J, et al. Disassembly sequence planning using discrete bees algorithm for human-robot collaboration in remanufacturing. Robot Comput-Integr Manuf 2020;62:101860.

[51] Belhadj I, Aicha M, Aifaoui N. Product disassembly planning and task allocation based on human and robot collaboration. Int J Interact Des Manuf 2022;16:803–19.

[52] Fang Y, Li Z, Wang S, Lu X. Multi-objective multi-fidelity optimisation for position-constrained human-robot collaborative disassembly planning. Int J Prod Res 2024;62(11):3872–89.

[53] Lee ML, Behdad S, Liang X, Zheng M. Disassembly sequence planning considering human-robot collaboration. Am Control Conf 2020;2020:2438–43.

[54] Guo XW, Fan CY, Zhou MC, et al. Human-robot collaborative disassembly line balancing problem with stochastic operation time and a solution via multi-objective shuffled frog leaping algorithm. IEEE Trans Autom Sci Eng 2023. https://doi.org/10.1109/TASE.2023.3296733.

[55] Li R, Pham DT, Huang J, et al. Unfastening of hexagonal headed screws by a collaborative robot. IEEE Trans Autom Sci Eng 2020;17:1455–68.

[56] Zhou Y, Peng Y, Li W, Pham DT. Stackelberg model-based human-robot collaboration in removing screws for product remanufacturing. Robot Comput-Integr Manuf 2022;77:102370.

[57] Liao H, Chen Y, Hu B, Behdad S. Optimization-based disassembly sequence planning under uncertainty for human-robot collaboration. J Mech Des 2023;145:22001.

[58] Yuan G, Liu X, Zhang C, Pham DT, Li Z. A new heuristic algorithm based on multi-criteria resilience assessment of human-robot collaboration disassembly for supporting spent lithium-ion battery recycling. Eng Appl Artif Intell 2023;126:106878.

[59] Fan J, Zheng P, Li S, Wang L. An integrated hand-object dense pose estimation approach with explicit occlusion awareness for human-robot collaborative disassembly. IEEE Trans Autom Sci Eng 2024;21:147–56.

[60] Cheng H., Xu W., Ai Q., et al. Manufacturing capability assessment for human-robot collaborative disassembly based on multi-data fusion. 45th SME North American Manufacturing Research Conference, 2017, 10: 26–36..

[61] Yuan G, Lv F, Shi J, et al. Integrated optimisation of human-robot collaborative disassembly planning and adaptive evaluation driven by a digital twin. Int J Prod Res 2024. https://doi.org/10.1080/00207543.2024.2381710.

[62] Qin SJ, Xiang CS, Wang JC, et al. Multiple product hybrid disassembly line balancing problem with human-robot collaboration. IEEE Trans Autom Sci Eng 2024. https://doi.org/10.1109/TASE.2024.3418898.

[63] Wang YQ, Li WD, Liang YC, Pham DT. Augmentation and classification of motor imagery electroencephalogram signals for human-robot collaborative disassembly. Int J Adv Manuf Technol 2024;133(11):5927–48.

[64] Huang J, Pham DT, Li R, et al. An experimental human-robot collaborative disassembly cell. Comput Ind Eng 2021;155:107189.

[65] Ding Y., Xu W., Liu Z., Zhou Z., Pham D.T.Robotic task oriented knowledge graph for human-robot collaboration in disassembly. 11th CIRP Conference on Industrial Product-Service Systems, 2019, 83: 105–110..

[66] Prioli JPJ, Rickli JL. Human-robot interaction for extraction of robotic disassembly information. Int J Comput Integr Manuf 2023:1–16.

[67] Fan J., Zheng P., Lee C.K.M.A multi-granularity scene segmentation network for human-robot collaboration environment perception. 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, Kyoto, Japan, 2022: 2105–2110..

[68] Liu Q, Liu Z, Xu W, et al. Human-robot collaboration in disassembly for sustainable manufacturing. Int J Prod Res 2019;57:4027–44.

[69] Zhang X, Yi D, Behdad S, Saxena S. Unsupervised human activity recognition learning for disassembly tasks. IEEE Trans Ind Inform 2024;20(1):785–94.

[70] Li S, You Y, Zheng P, et al. Mutual-cognition for proactive human-robot collaboration: a mixed reality-enabled visual reasoning-based method. IISE Trans 2024:1–13.

[71] Lou SH, Tan RJ, Zhang YR, et al. Personalized disassembly sequence planning for a human-robot hybrid disassembly cell. IEEE Trans Ind Inform 2024. https://doi.org/10.1109/TII.2024.3403254.

[72] Deng W, Liu Q, Zhao F, et al. Learning by doing: A dual-loop implementation architecture of deep active learning and human-machine collaboration for smart robot vision. Robot Comput-Integr Manuf 2024;86:102673.

[73] Guo L, Zhang ZQ, Wu TF, et al. Green and efficient-oriented human-robot hybrid partial destructive disassembly line balancing problem from non-disassemblability of components and noise pollution. Robot Comput Integr Manuf 2024;90:102816.

[74] Liu W, Liang X, Zheng M. Task-constrained motion planning considering uncertainty-informed human motion prediction for human–robot collaborative disassembly. IEEE/ASME Trans Mechatron 2023;28(4):2056–63.
[75] Jiao J, Feng GS, Yuan G. Research on the human-robot collaborative disassembly line balancing of spent lithium batteries with a human factor load. Batteries 2024; 10. https://doi.org/10.3390/batteries10060196.
[76] Zhang YJ, Liu L, Huang N, Radwin R, Li J. From manual operation to collaborative robot assembly: an integrated model of productivity and ergonomic performance. IEEE Robot Autom Lett 2021;6(2):895–902.
[77] Villani V, Pini F, Leali F, Secchi C. Survey on human-robot collaboration in industrial settings: safety, intuitive interfaces and applications. Mechatronics 2018;55:248–66.
[78] Slama I, Ben-Ammar O, Thevenin S, et al. Stochastic program for disassembly lot-sizing under uncertain component refurbishing lead times. Eur J Oper Res 2022; 303(3):1183–98.
[79] Xiao JH, Anwer N, Li WD, Eynard B, Zheng C. Dynamic Bayesian network-based disassembly sequencing optimization for electric vehicle battery. J Manuf Sci Technol 2022;38:824–35.
[80] Han M, Yun L, Li L. Deep reinforcement learning-based approach for dynamic disassembly scheduling of end-of-life products with stimuli-activated self-disassembly. J Clean Prod 2023;423:138758.
[81] Fang Y, Liu F, Li M, Cui H. Domain generalization-based dynamic multiobjective optimization: a case study on disassembly line balancing. IEEE Trans Evolut Comput 2023;27(6):1851–65.
[82] Zhou Y, Guo X, Li D. Correction to: a dynamic programming approach to a multi-objective disassembly line balancing problem. Ann Oper Res 2024;332:1249.
[83] Hartono N, Pham DT. A novel Fibonacci-inspired enhancement of the bees algorithm: application to robotic disassembly sequence planning. Cogent Eng 2024;11(1):1–27.
[84] Zhang X, Tian G, Fathollahi-Fard AM, et al. A chance-constraint programming approach for a disassembly line balancing problem under uncertainty. J Manuf Syst 2024;74:346–66.
[85] Qin S, Li C, Wang J, et al. Multiobjective human–robot collaborative disassembly sequence planning: considering the properties of components. IEEE Syst, Man, Cybern Mag 2024;10(2):15–23.
[86] Wang K, Guo J, Du B, et al. A novel MILP model and an improved genetic algorithm for disassembly line balancing and sequence planning with partial destructive mode. Comput Ind Eng 2023;186:109704.
[87] Cui J, Yang C, Zhang JL, et al. Robotic disassembly sequence planning considering parts failure features. IET Collab Intell Manuf 2023;5(1):12074.
[88] Hartono N, Ramírez FJ, Pham DT. A multiobjective decision-making approach for modelling and planning economically and environmentally sustainable robotic disassembly for remanufacturing. Comput Ind Eng 2023;184:109535.
[89] Sun X, Guo S, Guo J, Du B. A two-stage interactive evolutionary algorithm for multi-objective asynchronous parallel selective disassembly sequence planning problem. Comput Ind Eng 2023;175:108855.
[90] Zhang W, Zheng Y, Ahmad R. An energy-efficient multi-objective integrated process planning and scheduling for a flexible job-shop-type remanufacturing system. Adv Eng Inform 2023;56:102010.
[91] Kose Y, Ayyildiz E, Cevikcan E. Evaluation of disassembly line layouts using an integrated fermatean fuzzy decision-making methodology: an application for refrigerator disassembly line. Comput Ind Eng 2024;190:110090.
[92] Ren Y, Gao K, Fu Y, et al. Ensemble artificial bee colony algorithm with Q-learning for scheduling bi-objective disassembly line. Appl Soft Comput 2024; 155:111415.
[93] Hu P, Chu F, Dolgui A, Chu C, Liu M. Integrated multi-product reverse supply chain design and disassembly line balancing under uncertainty. Omega 2024;126: 103062.
[94] Ehm F. Scheduling and process planning for the dismantling shop with flexible disassembly mode and recovery level. Comput Ind Eng 2024;189:109927.
[95] Kalaycilar EG, Azizoğlu M, Batun S. Disassembly line balancing with hazardous task failures Model based solution approaches. Comput Ind Eng 2024;190: 110089.
[96] Guo L, Zhang Z, Wu T, et al. A grey wolf optimization algorithm for solving partial destructive disassembly line balancing problem consider feasibility evaluation and noise pollution. Adv Eng Inform 2024;60:102418.
[97] Lei S, Sun W, Yang Y. Comprehensive technology for recycling and regenerating materials from spent lithium iron phosphate battery. Environ Sci Technol 2024; 58(8):3609–28.
[98] Choux M, Pripp SW, Kvalnes F, Hellström M. To shred or to disassemble A techno-economic assessment of automated disassembly vs. shredding in lithium-ion battery module recycling. Resour, Conserv Recycl 2024;203:107430.
[99] Lu Y, Pei W, Peng K. State of the art of automatic disassembly of WEEE and perspective towards intelligent recycling in the era of Industry 4.0. Int J Adv Manuf Technol 2023;128:2825–43.
[100] Zeng Y, Zhang Z, Yin T, Zheng H. Robotic disassembly line balancing and sequencing problem considering energy-saving and high-profit for waste household appliances. J Clean Prod 2022;381(1):135209.
[101] Foo G, Kara S, Pagnucco M. Artificial learning for part identification in robotic disassembly through automatic rule generation in an ontology. IEEE Trans Autom Sci Eng 2023;20(1):269–309.
[102] Zafar MH, Langas EF, Sanfilippo F. Exploring the synergies between collaborative robotics, digital twins, augmentation, and industry 5.0 for smart manufacturing: a state-of-the-art review. Robot Comput-Integr Manuf 2024;89:102769.
[103] Li C, Zheng P, Zhou P, Yin Y, Lee C, Wang L. Unleashing mixed-reality capability in deep reinforcement learning-based robot motion generation towards safe human–robot collaboration. J Manuf Syst 2024;74:411–21.
[104] Ciccarelli M, Papetti A, Germani M. Exploring how new industrial paradigms affect the workforce: a literature review of operator 4.0. J Manuf Syst 2023;70: 464–83.
[105] Bitencourt J, Wooley A, Harris G. Verification and validation of digital twins: a systematic literature review for manufacturing applications. Int J Prod Res 2024; 01:1–29.
[106] Liu B, Zhang Y, Sheng G, et al. Behavior-environment interaction aware manufacturing service collaboration optimization. J Manuf Syst 2024;74:302–15.
[107] Maurtua I, Ibarguren A, Kildal J, Susperregi L, Sierra B. Human-robot collaboration in industrial applications: safety, interaction and trust. Int J Adv Robot Syst 2017;14(4).
[108] Inkulu AK, Bahubalendruni MVAR, Dara AKS. Challenges and opportunities in human robot collaboration context of Industry 4.0 - a state of the art review. Ind Robot Int J Robot Res Appl 2022;49(2):226–39.
[109] Wang L, Gao R, Váncza J, Krüger J, et al. Symbiotic human-robot collaborative assembly. CIRP Ann Manuf Technol 2019;68(2):701–26.
[110] Gervasi R, Capponi M, Mastrogiacomo L, Franceschini F. Manual assembly and Human-robot collaboration in repetitive assembly processes: a structured comparison based on human-centered performances. Int J Adv Manuf Technol 2023;126:1213–31.
[111] Verna E, Puttero S, Genta G, Galetto M. Exploring the effects of perceived complexity criteria on performance measures of human-robot collaborative assembly. J Manuf Sci Eng 2023;145(10):101014.
[112] Awolusi I, Nnaji C, Okpala I, Albert A. Adaptation behavior of construction workers using wearable sensing devices for safety and health monitoring. J Manag Eng 2024;40(1):04023055.
[113] Wu H, Li S, Wu D. Motor imagery classification for asynchronous EEG-based brain-computer interfaces. IEEE Trans Neural Syst Rehabil Eng 2024;32:527–36.
[114] Eswaran M, Gulivindala AK, Inkulu AK, et al. Augmented reality-based guidance in product assembly and maintenance/repair perspective: a state of the art review on challenges and opportunities. Expert Syst Appl 2023;213:118983.
[115] Coronado E, Kiyokawa T, Gustavo A, et al. Evaluating quality in human-robot interaction: a systematic search and classification of performance and human-centered factors, measures and metrics towards an industry 5.0. J Manuf Syst 2022;63:392–410.
[116] Zhang X, Eltouny K, Liang X, Behdad S. Automatic screw detection and tool recommendation system for robotic disassembly. ASME J Manuf Sci Eng 2023; 145(3):031008.
[117] Chu H, Zhang T, Zou Y, Sun H. Follower to collaborator: a robot proactive collaborative controller based on human multimodal information for 3D handling/assembly scenarios. J Manuf Syst 2024;73:287–306.
[118] Mao Z, Sun Y, Fang K, et al. Model and metaheuristic for human–robot collaboration assembly line worker assignment and balancing problem. Comput Oper Res 2024;165:106605.
[119] Peng Y, Li W, Liang Y, Pham DT. Robotic disassembly of screws for end-of-life product remanufacturing enabled by deep reinforcement learning. J Clean Prod 2024;439:140863.
[120] Bharti V, Kumar A, Purohit V, et al. A label-efficient semi self-supervised learning framework for IoT devices in industrial process. IEEE Trans Ind Inform 2024;20 (2):2253–62.
[121] Sardar SK, Lim CH, Yoon SH, Lee SC. Ergonomic risk assessment of manufacturing works in virtual reality context. Int J Hum-Comput Interact 2023:1–17.
[122] Zhang Y, Hang JL, Zhu TQ, et al. Functional grasp: learning functional grasp for robots via semantic hand-object representation. IEEE Robot Autom Lett 2023;8 (5):3094–101.
[123] Marti-Puig P, Touhami IA, Perarnau RC, Serra-Serra M. Industrial AI in condition-based maintenance: a case study in wooden piece manufacturing. Comput Ind Eng 2024;188:109907.
[124] Yuan G, Liu X, Zhu C, et al. Multi-objective coupling optimization of electrical cable intelligent production line driven by digital twin. Robot Comput-Integr Manuf 2024;86:102682.
[125] Tozanlı Ö, Kongar E, Gupta SM. Trade-in-to-upgrade as a marketing strategy in disassembly-to-order systems at the edge of blockchain technology. Int J Prod Res 2020;58(23):7183–200.
[126] Wang C, Zhao J. Role dynamic assignment of human–robot collaboration based on target prediction and fuzzy inference. IEEE Trans Ind Inform 2024;20(1): 471–81.